

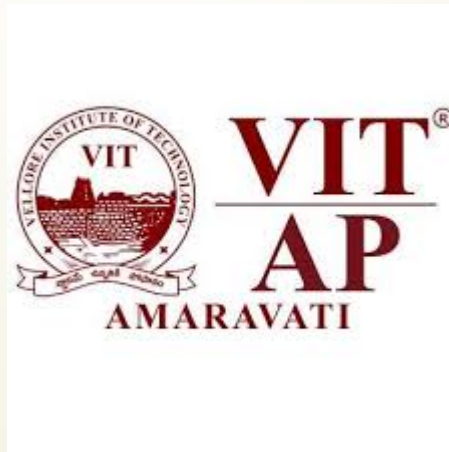
Combinational Logic Gates in CMOS

By

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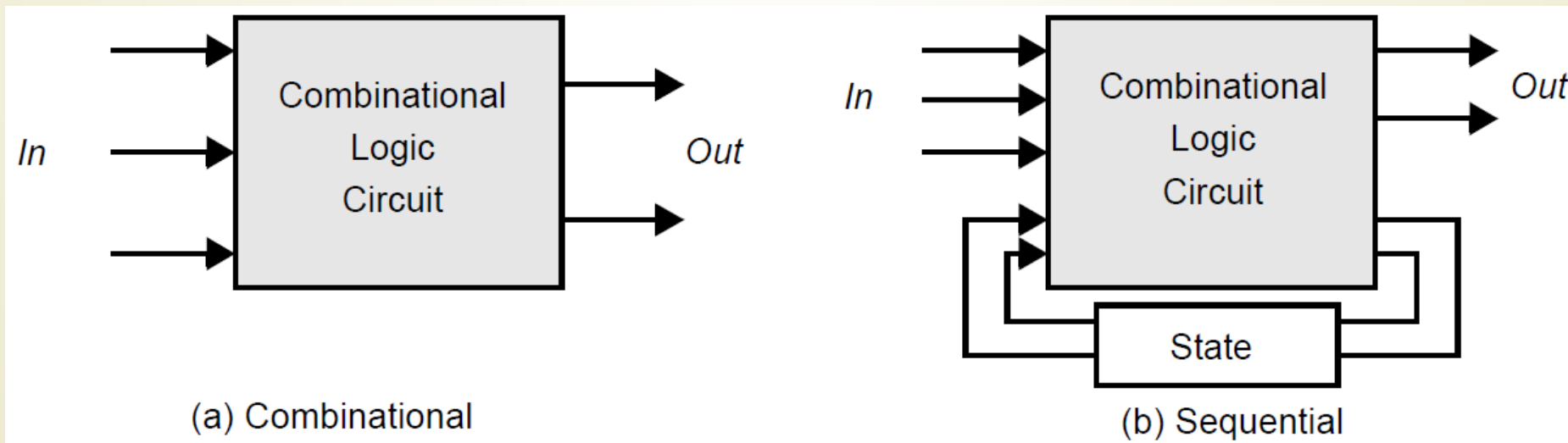
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Introduction

- ▶ The focus is on combinational logic (or non-regenerative) circuits; this is, circuits that have the property that at any point in time, the output of the circuit is related to its current input signals.
- ▶ This is in contrast to another class of circuits, known as sequential or regenerative, for which the output is not only a function of the current input data, but also of previous values of the input signals. This is accomplished by connecting one or more outputs intentionally back to some inputs. Consequently, the circuit “remembers” past events and has a sense of history.

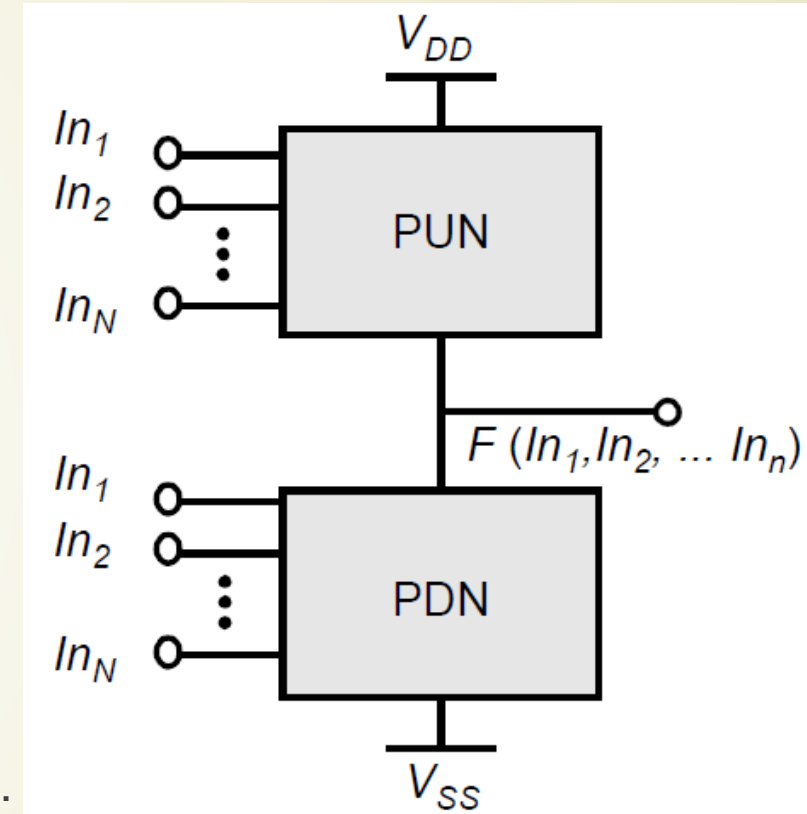


Static and Dynamic Logic Design

Static Design - At every point of time each gate output is either connected to V_{DD} or V_{SS} via a low resistance path. The static CMOS style is really an extension of the static CMOS inverter to multiple inputs.

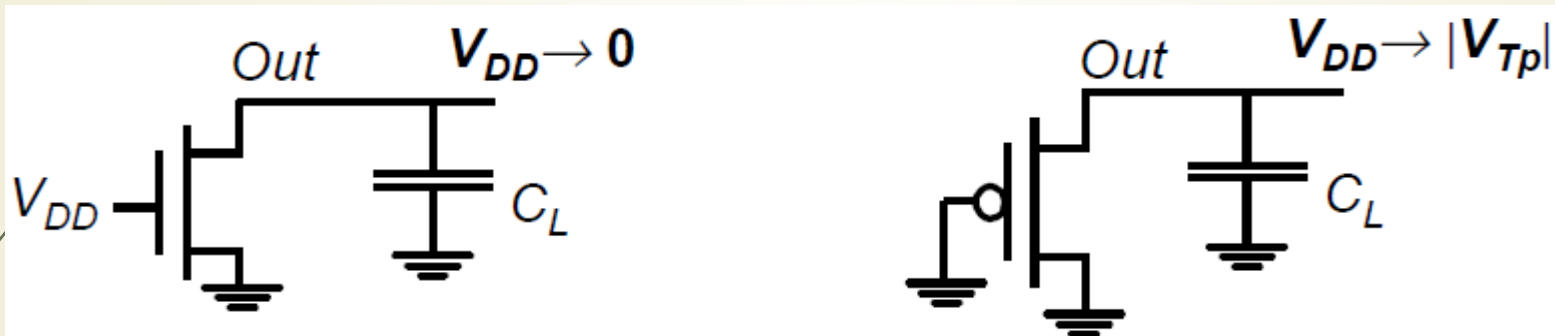
Dynamic Design - This relies on temporary storage of signal values on the capacitance of high-impedance circuit nodes.

Complementary CMOS - A static CMOS gate is a combination of two networks, called the pull-up network (PUN) and the pull-down network (PDN). The function of the PUN is to provide a connection between the output and V_{DD} . Similarly, the function of the PDN is to connect the output to V_{SS} .

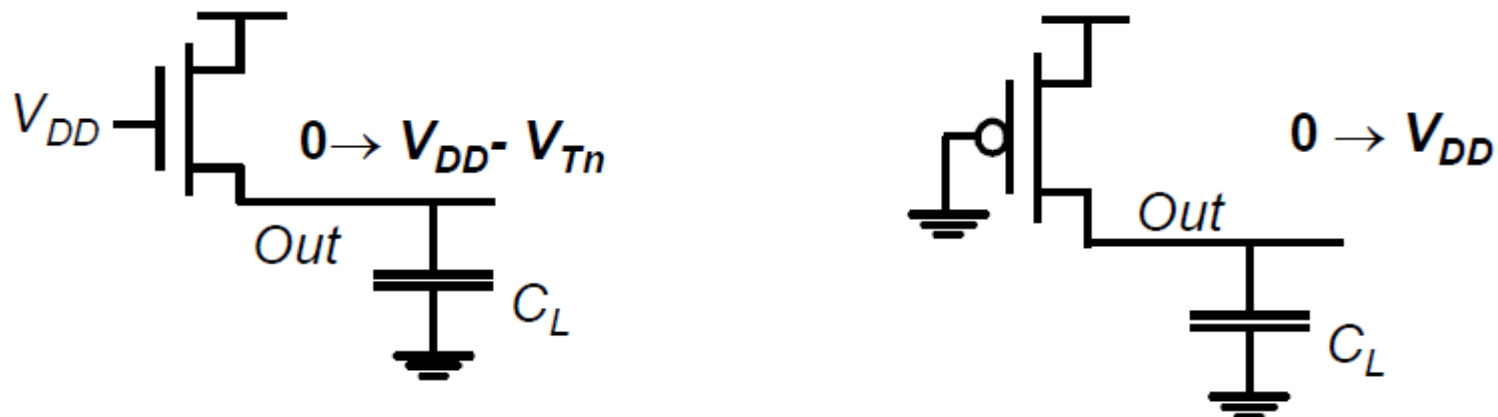


Choice of Network

- ▶ The PDN is constructed using NMOS devices, while PMOS transistors are used in the PUN. The primary reason for this choice is that NMOS transistors produce “strong zeros,” and PMOS devices generate “strong ones”. To illustrate this, consider the examples shown in Figure.



(a) pulling down a node using NMOS and PMOS switches



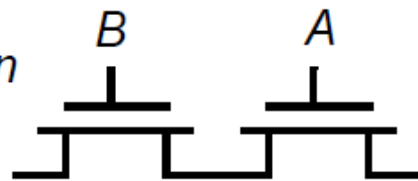
(b) pulling up a node using NMOS and PMOS switches

Logic Rules

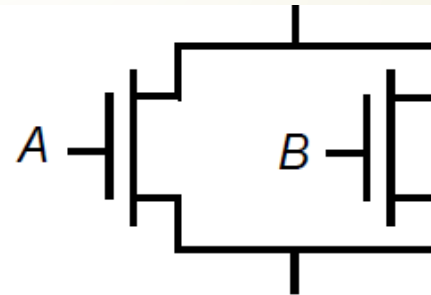
- ▶ A set of construction rules can be derived to construct logic functions NMOS devices connected in series corresponds to an AND function. With all the inputs high, the series combination conducts and the value at one end of the chain is transferred to the other end.
- ▶ Similarly, NMOS transistors connected in parallel represent an OR function. A conducting path exists between the output and input terminal if at least one of the inputs is high.
- ▶ A series connection of PMOS conducts if both inputs are low, representing a NOR function $(\overline{A} \cdot \overline{B} = \overline{A+B})$, while PMOS transistors in parallel implement a NAND $(\overline{A+B} = \overline{A} \cdot \overline{B})$.

Series Combination

Conducts if $A \cdot B$



(a) series



Parallel Combination
Conducts if $A + B$

(b) parallel

Problem - Two-input NAND Gate

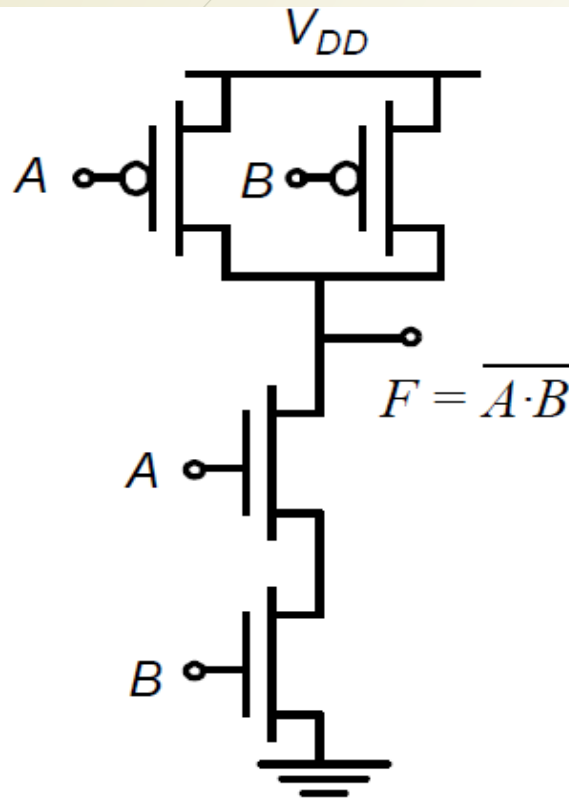


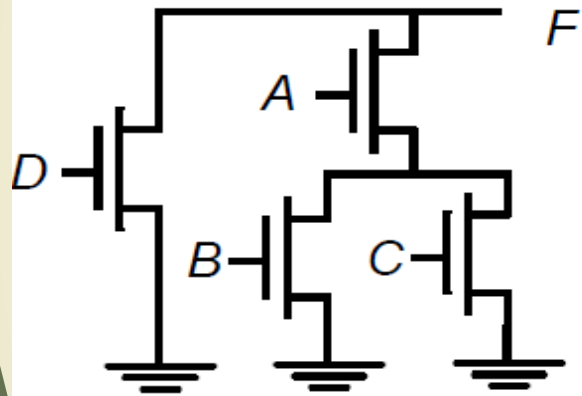
Table Truth Table for 2 input NAND

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

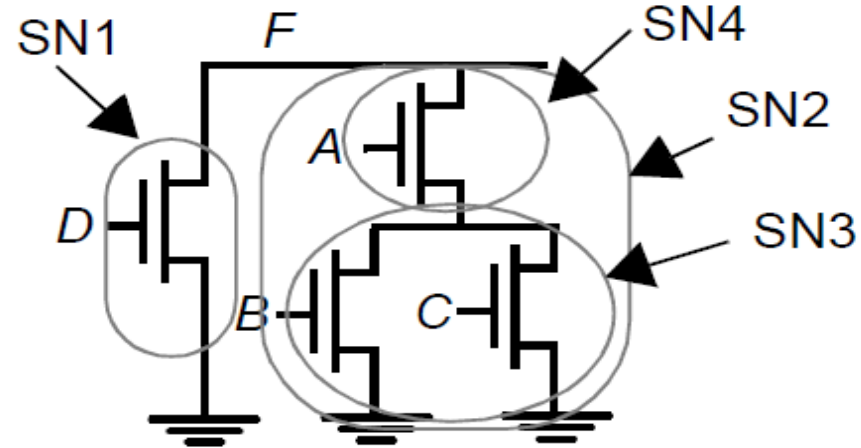
Figure Two-input NAND gate in complementary static CMOS style.

Synthesis of complex CMOS Gate

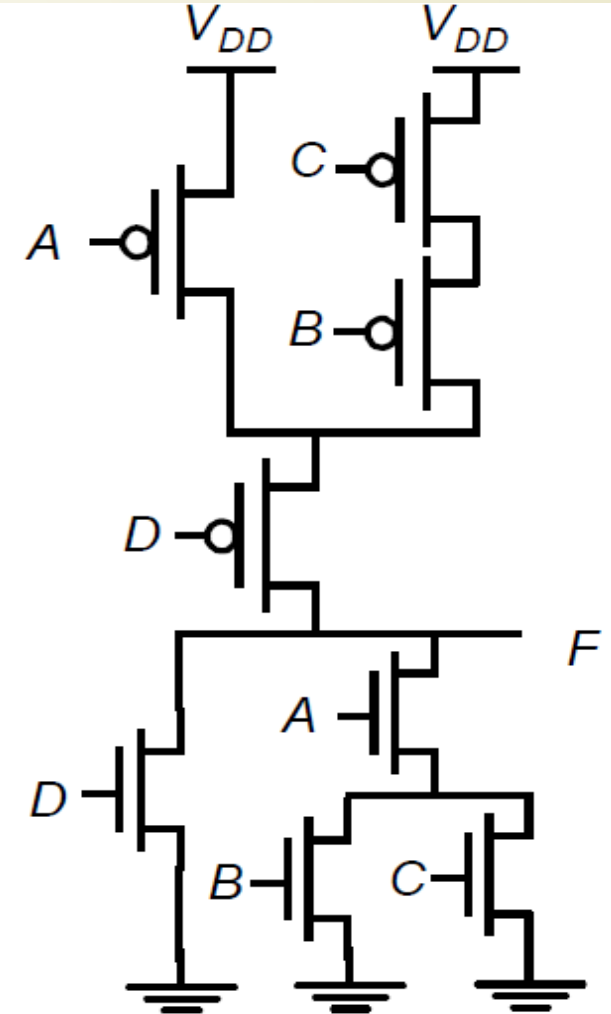
$$F = \overline{D + A \cdot (B + C)}$$



(a) pull-down network



(b) Deriving the pull-up network hierarchically by identifying sub-nets



(c) complete gate

Figure Complex complementary CMOS gate.

Static Properties of Complementary CMOS Gates

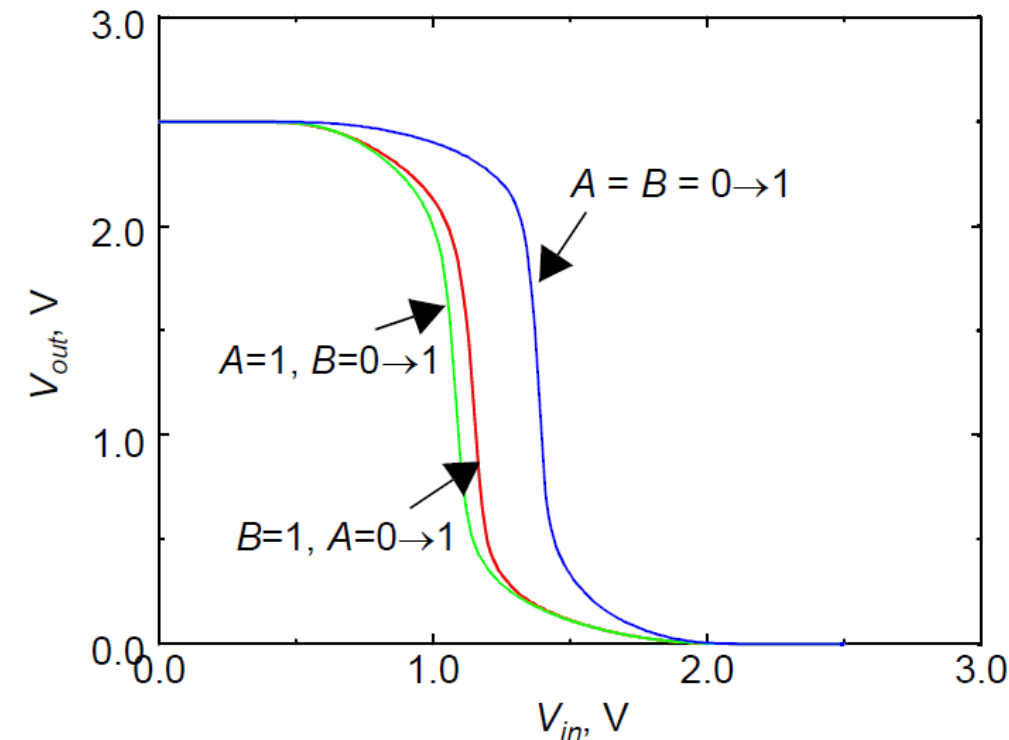
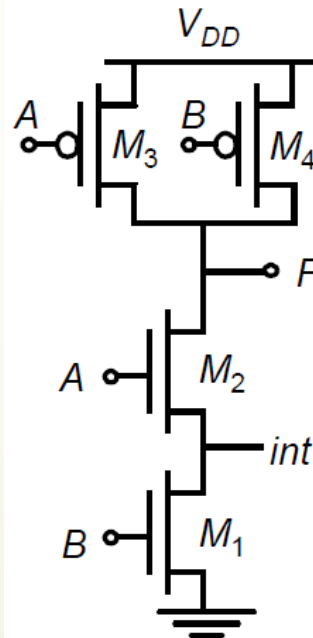
The analysis of the DC voltage transfer characteristics and the noise margins is more complicated than for the inverter, as these parameters depend upon the data input patterns applied to gate.

Consider three possible inputs in NAND gate. Consequently, voltage transfer curves display significant differences. The large variation between case (a) and the others (b & c) is explained by the fact. The VTC is shifted to the left as a result of the weaker PUN.

The difference between (b) and (c) results mainly from the state of the internal node *int* between the two NMOS devices.

The NMOS devices to turn on, both gate-to-source voltages must be above V_{Tn} with $V_{GS2} = V_A - V_{DS1}$ and $V_{GS1} = V_B$.

The threshold voltage of M_2 will be higher than M_1 due to body effect.



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Figure The VTC of a two-input NAND is data-dependent.

continue.....

The threshold voltages of the two devices are given by

$$V_{Tn2} = V_{tn0} + \gamma((\sqrt{|2\phi_f|} + V_{int}) - \sqrt{|2\phi_f|})$$

$$V_{Tn1} = V_{tn0}$$

For case (b), M_3 is turned off, and the gate voltage of M_2 is set to V_{DD} . Since the drive on M_2 is large, this resistance is small and has only a small effect on the voltage transfer characteristics. In case (c), transistor $M1$ acts as a resistor, causing body effect in $M2$. The overall impact is quite small as seen from the plot.

There are some conclusion from above analysis

- Propagation delay is input dependent.
- Noise margin is input dependent
- **PUN** is stronger means **NM_H** is stronger.
- **PDN** is stronger means **NM_L** is stronger.

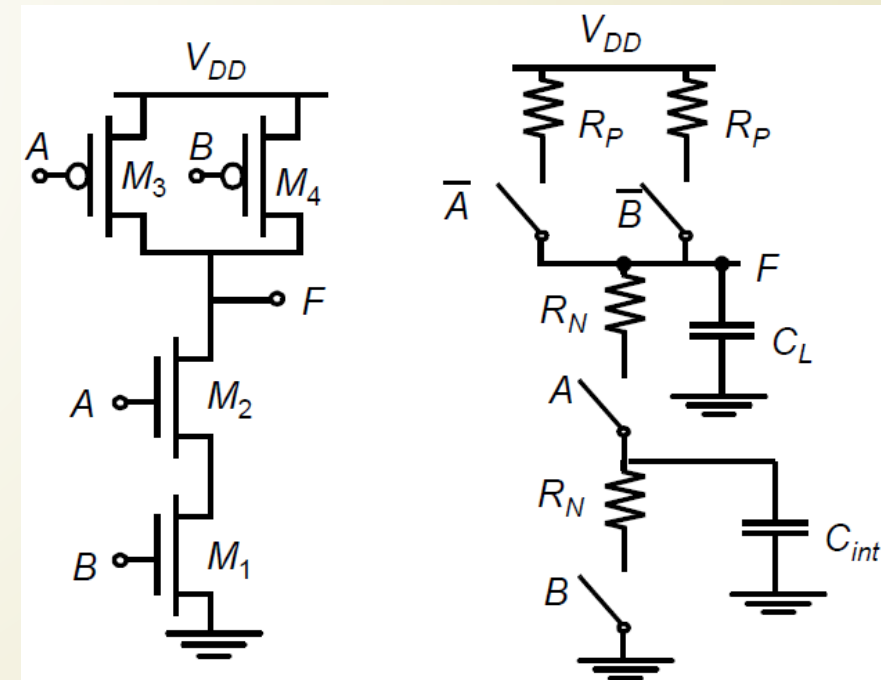
Propagation Delay of Complementary CMOS Gates

The delay can be computed by considering each transistor is modeled as a resistor in series with an ideal switch. The logic is transformed into an equivalent RC network that includes the effect of internal node capacitances. The propagation delay depends upon the input patterns.

If both inputs are driven low, the two PMOS devices are ON. The delay in this case is $0.69 \times (R_p/2) \times C_L$, since the two resistors are in parallel. The worst case occur when only one device is ON - $0.69 \times R_p \times C_L$.

The output is discharged only if both A and B are switched high, and the delay is given by $0.69 \times 2R_N \times C_L$. In other words, adding devices in series slows down the circuit, and devices must be made wider to avoid a performance penalty.

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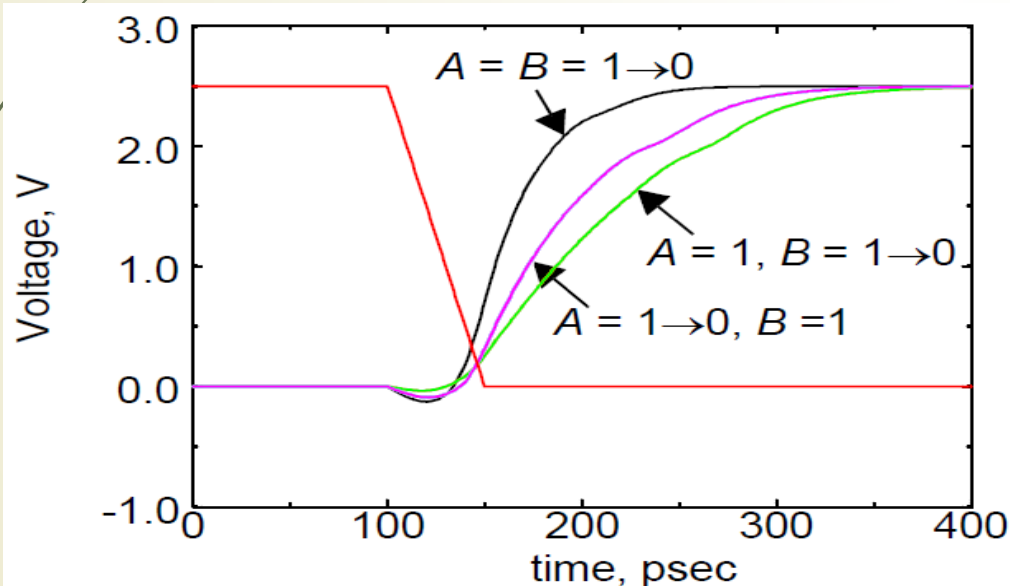
(a) Two-input NAND

(b) RC equivalent model

Delay dependence on input patterns

As expected, the case where both inputs transition go low ($A = B = 1 \rightarrow 0$) results in a smaller delay, compared to the case where only one input is driven low.

- Notice that the worst-case low-to-high delay depends upon which input (A or B) goes low. For the case that $B = 1$ and A transitions from $1 \rightarrow 0$, the pull-up PMOS device only has to charge up the output node capacitance since M2 is turned off.
- On the other hand, for the case where $A=1$ and B transitions from $1 \rightarrow 0$, the pull-up PMOS device has to charge up the sum of the output and the internal node capacitances, which slows down the transition



Input Data Pattern	Delay (psec)
$A=B=1 \rightarrow 0$	35
$A=1, B=1 \rightarrow 0$	76
$A=1 \rightarrow 0, B=1$	57

Figure Example showing the delay dependence on input patterns.

CMOS implementation of NOR gate

- ▶ In case of NOR gate output of the network is high if and only if both inputs A and B are low. The worst-case pull-down transition happens when only one of the NMOS devices turns ON (if either A or B is high).
- ▶ Assume that the goal is to size the NOR gate such that it has approximately the same delay as an inverter with the following device sizes, NMOS $0.5\mu\text{m}/0.25\mu\text{m}$ and PMOS $1.5\mu\text{m}/0.25\mu\text{m}$.
- ▶ For the output to be pulled high, both devices must be turned on
- ▶ However, in more complex logic gates that have large
- ▶ fan-in, the internal node capacitances can become
- ▶ significant

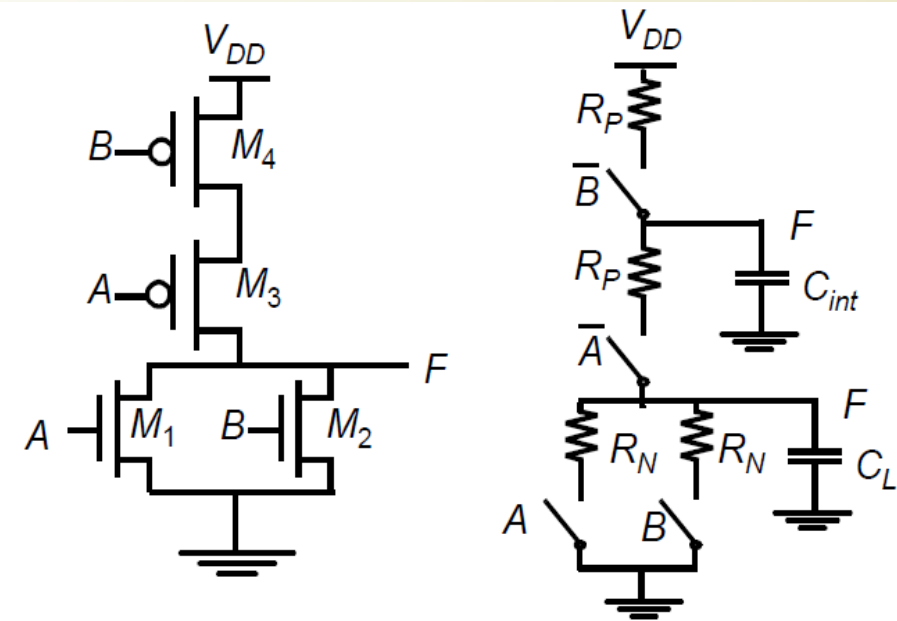


Figure Sizing of a NOR gate to produce the same delay as an inverter with size of NMOS: $0.5\mu\text{m}/0.25\mu\text{m}$ and PMOS: $1.5\mu\text{m}/0.25\mu\text{m}$.

Continue....

- ▶ The switching threshold, V_M , is defined as the point where $V_{in} = V_{out}$. Its value can be obtained graphically from the intersection of the VTC with the line given by $V_{in} = V_{out}$. In this region, both PMOS and NMOS are always saturated.
- ▶ Consider a 4-input NAND gate as shown in Figure which shows the equivalent RC model of the gate, including the internal node capacitances.
- ▶ The delay analysis for such a circuit involves
- ▶ Solving distributed RC networks.
- ▶ Consider the pull-down delay of the circuit.
- ▶ The output is discharged when all inputs are driven high.

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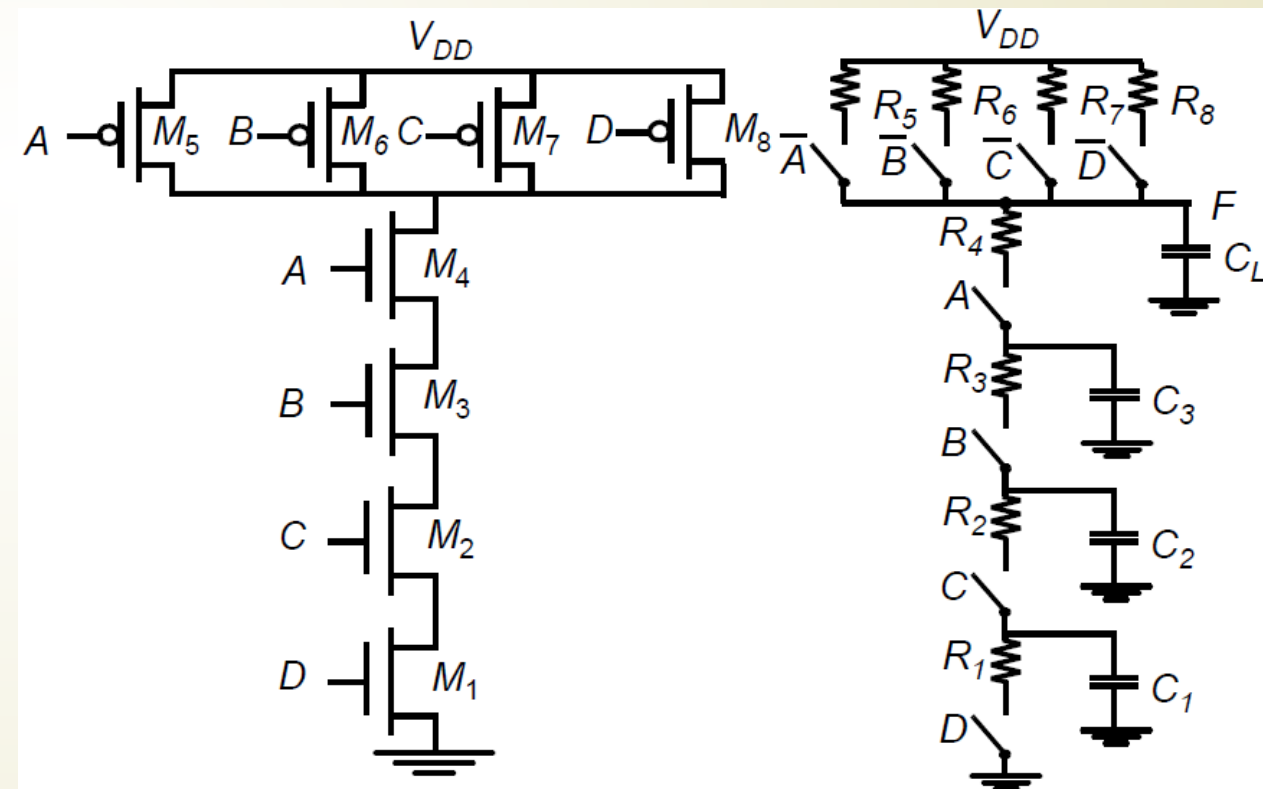


Figure Four input NAND gate and its RC model

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- ▶ The propagation delay can be computed using the Elmore delay model and is approximated as:

$$t_{pHL} = 0.69(R_1 \cdot C_1 + (R_1 + R_2) \cdot C_2 + (R_1 + R_2 + R_3) \cdot C_3 + (R_1 + R_2 + R_3 + R_4) \cdot C_L) \quad \text{.....(1)}$$

- ▶ Assuming that all NMOS devices have an equal size

$$t_{pHL} = 0.69R_N(C_1 + 2 \cdot C_2 + 3 \cdot C_3 + 4 \cdot C_L) \quad \text{.....(2)}$$

Problems in CMOS

- complementary CMOS is a very robust and simple approach for implementing logic gates.
- First, the number of transistors required to implement an N fan-in gate is $2N$. This can result in significant implementation area.
- The large number of transistors ($2N$) increases the overall capacitance of the gate.
- Propagation delay of the gates deteriorate as a function of fan-in.
- The series connection of the transistor in either PUN or PDN network causes an additional slowdown.
- Therefore the delay becomes a quadratic function of fan-in.
- For an N -input NAND gate, the output capacitance increases linearly with the fan-in since the number of PMOS devices connected to the output node increases linearly with the fan-in. It also includes series connection of transistor because the effective (dis)charging resistance is increased. So the overall high-to-low delay can increase in a quadratic fashion.

Propagation delay as a function of fan-in

- ▶ The t_{pLH} increases linearly due to the linearly-increasing value of the output capacitance.
- ▶ The simultaneous increase in the pull-down resistance and the load capacitance results in an approximately quadratic relationship for t_{pHL} .

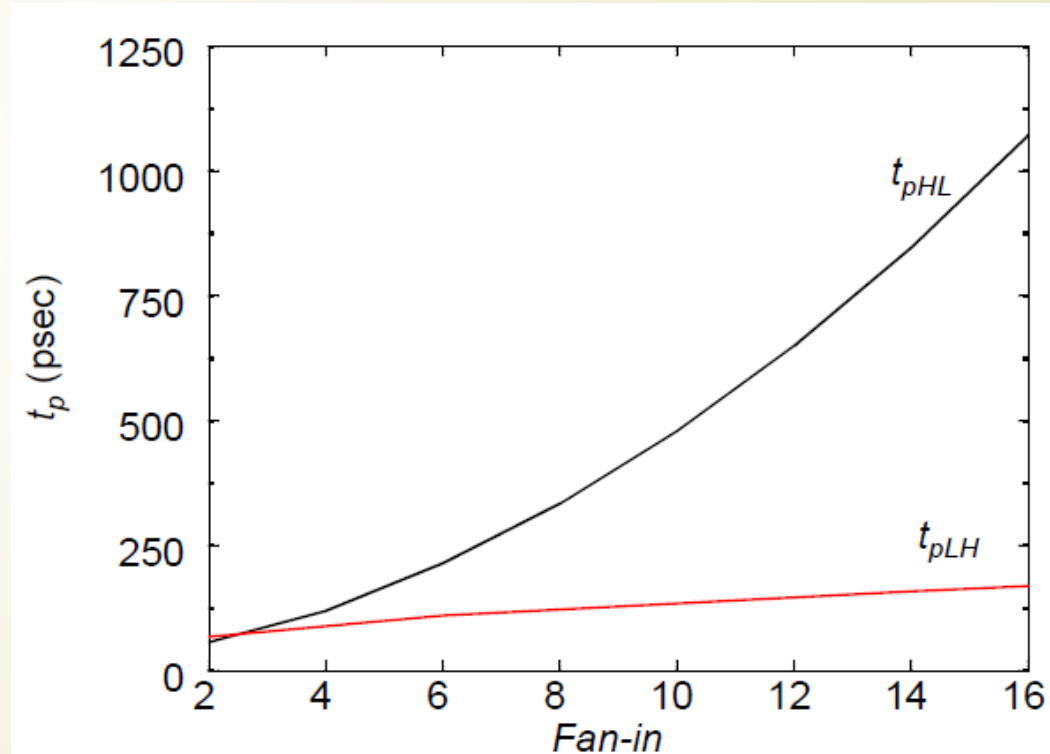
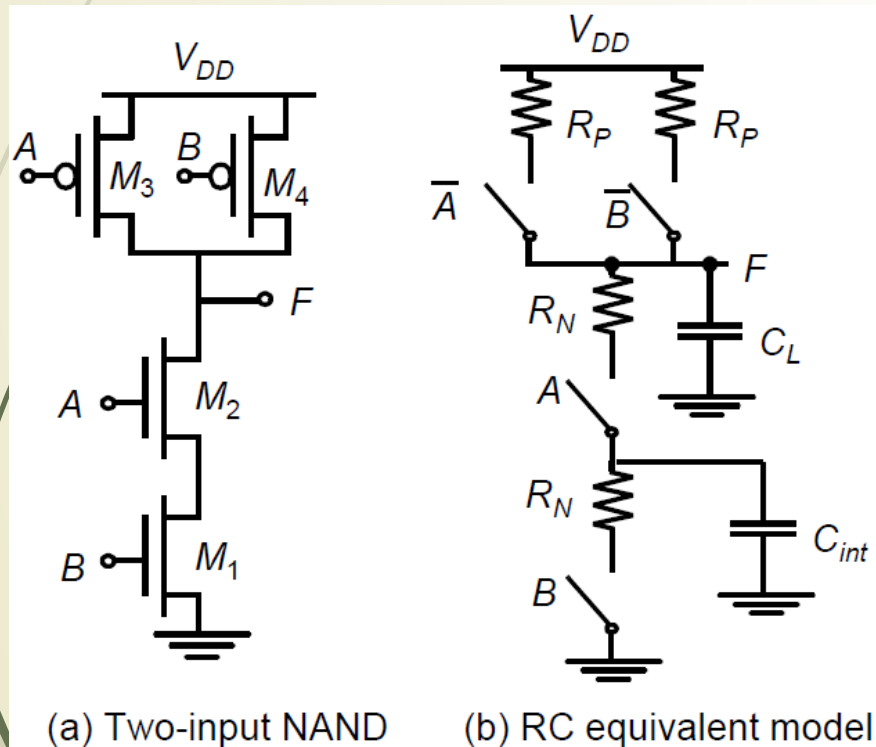


Figure Propagation delay of CMOS NAND gate as a function of fan-in.

Design Techniques for large Fan-in

Several approaches may be used to reduce delays in large fan-in circuits.

Transistor Sizing -

- The most obvious solution is to increase the overall transistor size. This lowers the resistance of devices in series and lowers the time constant. However, increasing the transistor size, results in larger parasitic capacitors, which do not only affect the propagation delay of the gate in question. but also present a larger load to the preceding gate.
- This technique should, therefore, be used with caution. If the load capacitance is dominated by the intrinsic capacitance of the gate, widening the device only creates a “self-loading” effect, and the propagation delay is unaffected.

Progressive Transistor Sizing

- ▶ An alternate approach to uniform sizing is to use progressive transistor sizing –
- ▶ Referring equation 1 we see that the resistance of M_1 (R_1) appears N times (4 times) in the delay equation. the resistance of M_2 (R_2) appears $N-1$ times (3 times), etc.
- ▶ From the equation, it is clear that R_1 should be made the smallest, R_2 the next smallest, etc.
- ▶ Consequently, a progressive scaling of the transistors is beneficial: $M_1 > M_2 > M_3 > M_N$.
- ▶ Basically, in this approach, the important resistance is reduced while reducing capacitance.

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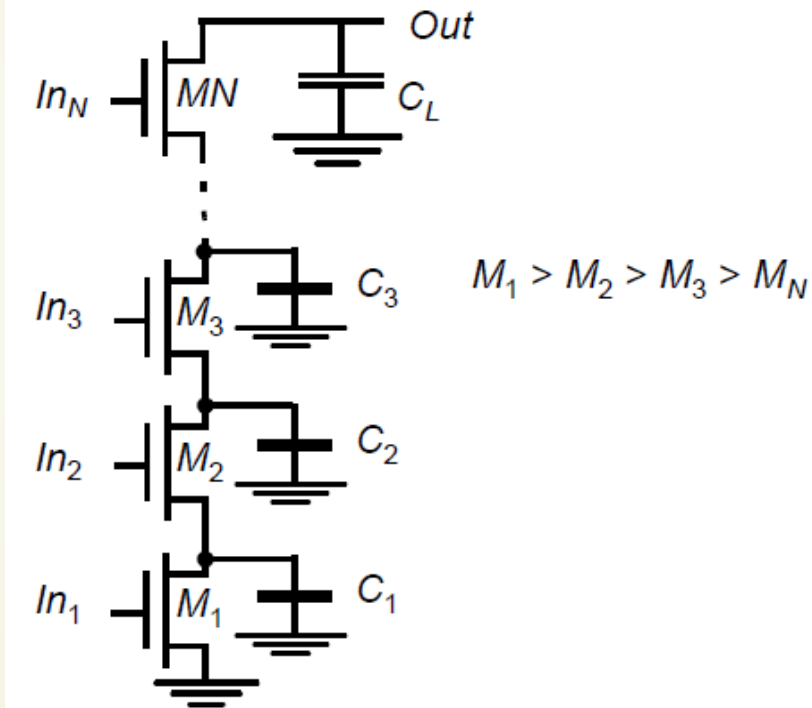


Figure Progressive sizing of transistors in large transistor chains copes with the extra load of internal capacitances.

Input Re-Ordering

- An input signal to a gate is called critical if it is the last signal of all inputs to assume a stable value. The path through the logic which determines the ultimate speed of the structure is called the critical path.
- Putting the critical-path transistors closer to the output of the gate can result in a speedup. Assume also that C_L is initially charged high. In case (a), no path to GND exists until M_1 is turned ON.
- The delay between the arrival of In_1 and the output is therefore determined by the time it takes to discharge C_L , C_1 and C_2 .
- In the second case, C_1 and C_2 are already discharged when In_1 changes. Only C_L still has to be discharged, resulting in a smaller delay.

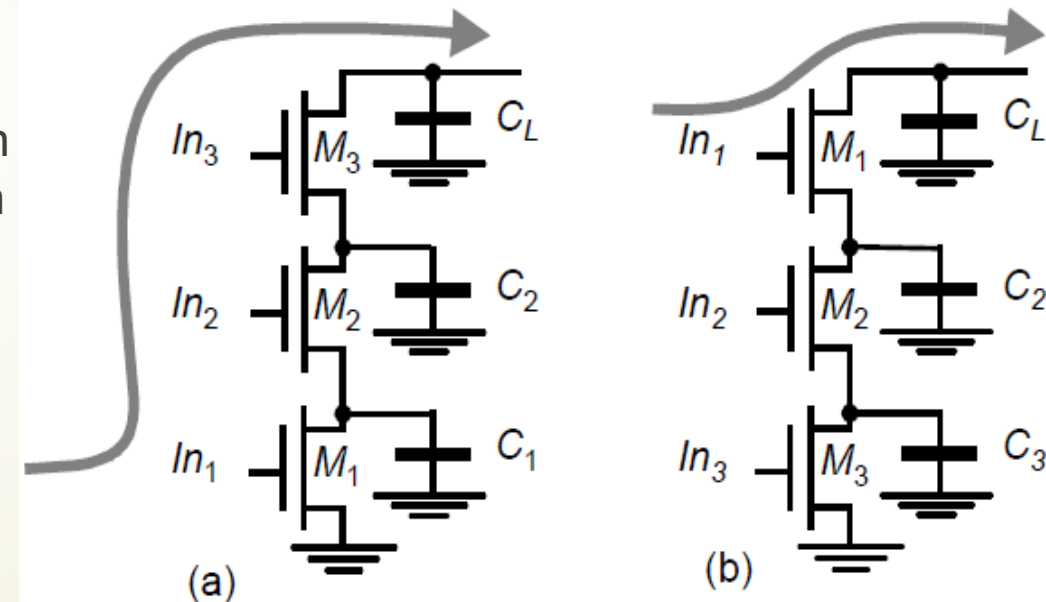


Figure Influence of transistor ordering on delay.
Signal In_1 is the critical signal.

Logic Restructuring

- ▶ Manipulating the logic equations can reduce the fan-in requirements and hence reduce the gate delay as shown in figure.
- ▶ It is advisable to have symmetric function to derive the signal at proper time.
- ▶ The quadratic dependency of the gate delay on *fan-in* makes the six-input NOR gate extremely slow.
- ▶ Partitioning the NOR-gate into two three input gates results in a significant speed-up, which offsets by far the extra delay.

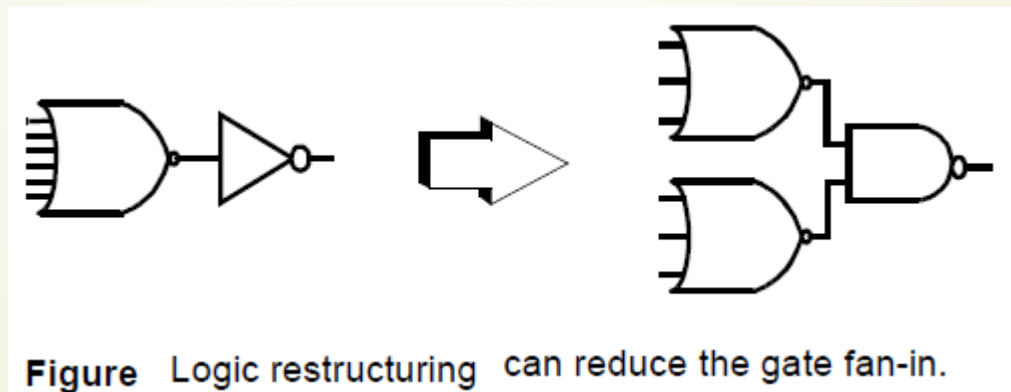


Figure Logic restructuring can reduce the gate fan-in.

Power Consumption in CMOS Logic Gates

- ▶ The dynamic power dissipation is given by

$$\alpha_{0 \rightarrow 1} C_L V_{DD}^2 f \quad \dots\dots\dots (3)$$

- ▶ Here $\alpha_{0 \rightarrow 1}$ is called the switching activity, which is having two components .
- ▶ 1- A static component that is only a function of the topology of the logic network.
- ▶ 2- Dynamic one that results from the timing behavior of the circuit—the latter factor is also called glitching.
- ▶ The probability P_0 that the output will be in the zero state in one cycle, multiplied by the probability P_1
- ▶ that the output will be in the one state in the next cycle

$$\alpha_{0 \rightarrow 1} = p_0 \bullet p_1 = p_0 \bullet (1 - p_0) \quad \dots\dots\dots (4)$$

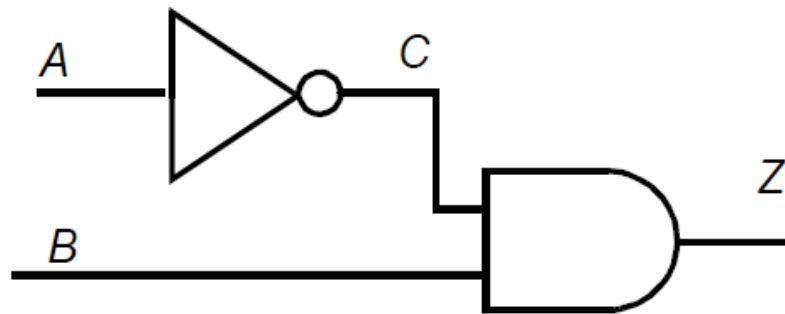
Inter-signal Correlations

- ▶ The probability that the node Z undergoes a power consuming transition is then determined using the AND-gate expression.

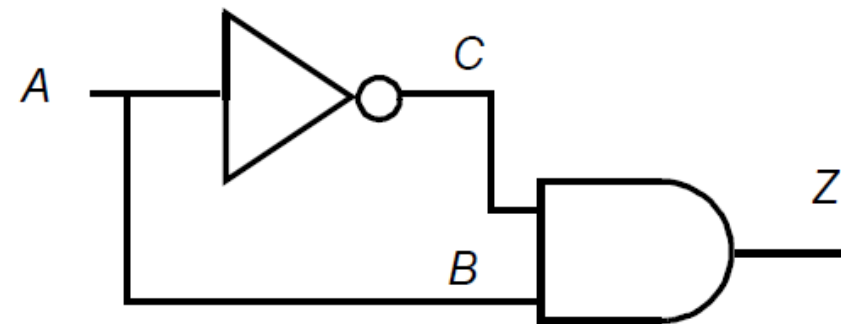
$$p_{0 \rightarrow 1} = (1 - p_a p_b) p_a p_b = (1 - 1/2 \cdot 1/2) 1/2 \cdot 1/2 = 3/16$$

- ▶ The first approach has two major limitations (1) it does not deal with circuits with feedback as found in sequential circuits (2) it assumes that the signal probabilities at the input of each gate are independent.
- ▶ If a dependency between the two exists a conditional probability has to be employed, such as

$$p_Z = p(C=1|B=1) \cdot p(B=1) \quad \dots\dots(3)$$



(a) Logic circuit without reconvergent fanout



(b) Logic circuit with reconvergent fanout

Figure Example illustrating the effect of signal correlations.

Dynamic or Glitching Transitions

- ▶ The finite propagation delay from one logic block to the next can cause spurious transitions, called glitches, critical races, or dynamic hazards.
- ▶ We can see in the example for this particular transition, all the odd bits must transition to 0 while the even bits remain at the value of 1. Due to the finite propagation delay, the higher order even outputs start to discharge and the voltage drops.
- ▶ Glitches contribute significantly to the power dissipation.
- ▶ Long chains of gates often occur in important and multipliers and the glitching component can easily dominate the overall power Consumption.

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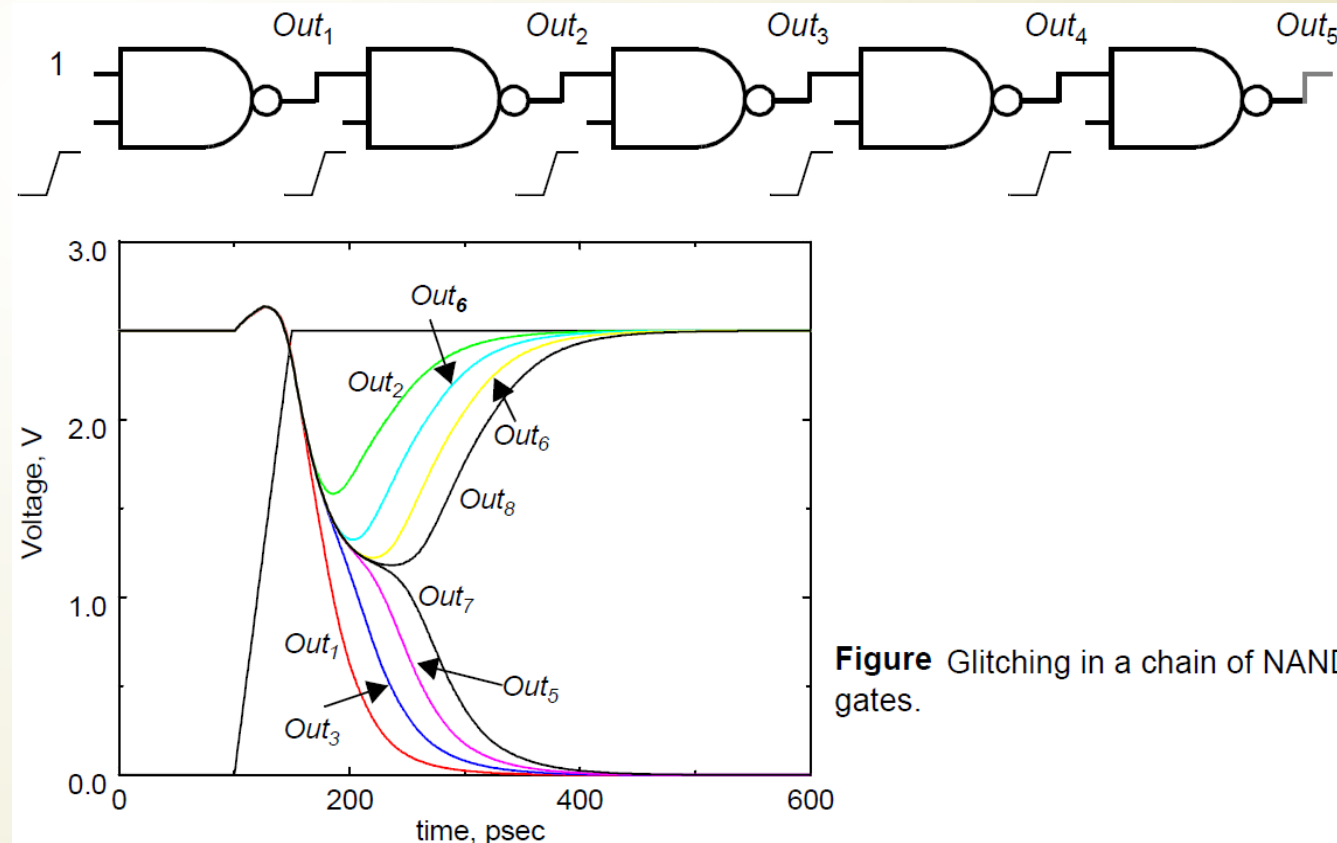
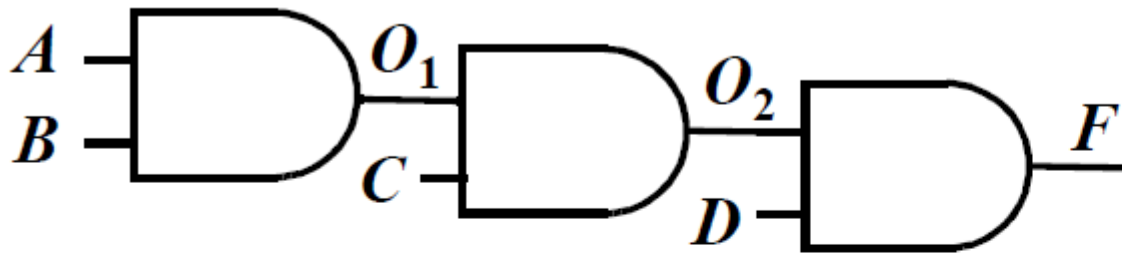


Figure Glitching in a chain of NAND gates.

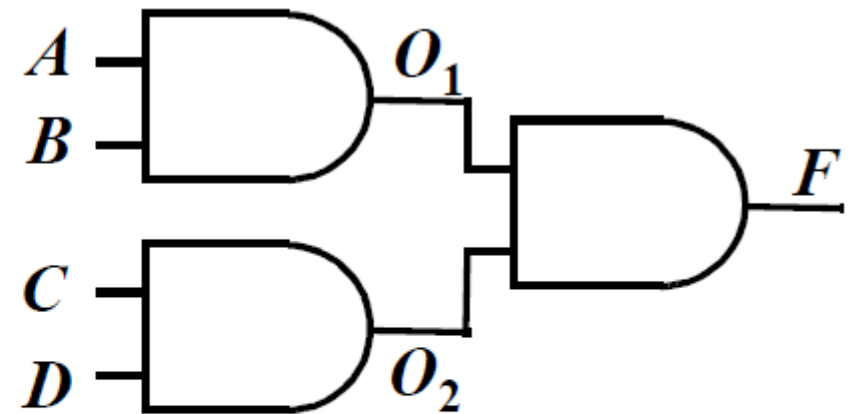
Design Techniques to Reduce Switching Activity

Logic Restructuring

- Changing the topology of a logic network may reduce its power dissipation. Consider for instance two alternate implementations of $F = A \cdot B \cdot C \cdot D$.
- In this example the tree topology will have lower (no) glitching activity since the signal paths are balanced to all the gates.



Chain structure

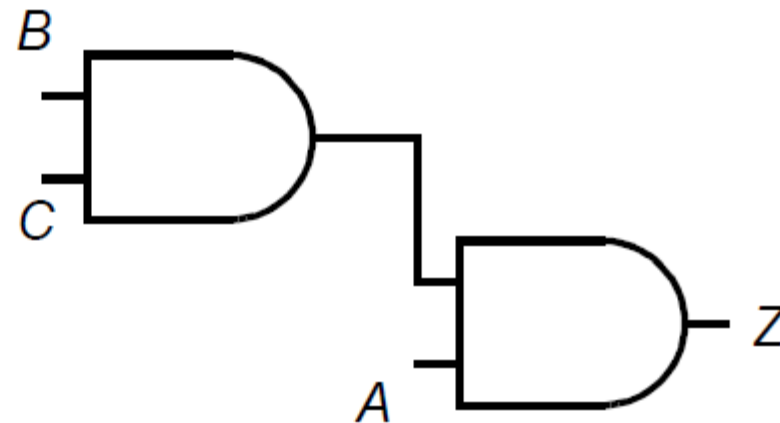
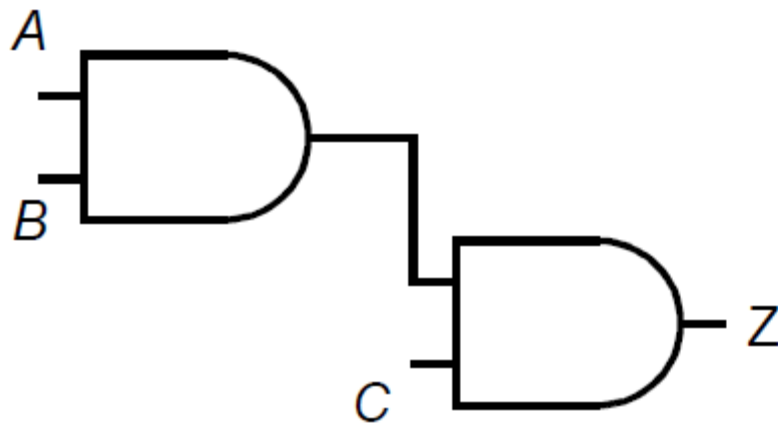


Tree structure

Figure Simple example to demonstrate the influence of circuit topology on activity.

Input ordering

- Consider the two static logic circuits of Figure. Both the circuits implement identical logic functionality and have equal activity at the output node.
- The difference is in the activity at the intermediate node. In the first circuit, this activity equals $(1 - 0.5 \times 0.2) (0.5 \times 0.2) = 0.09$.
- In the second case, the probability that a $0 \rightarrow 1$ transition occurs equals $(1 - 0.2 \times 0.1) (0.2 \times 0.1) = 0.0196$.



$$P_{(A=1)} = 0.5$$

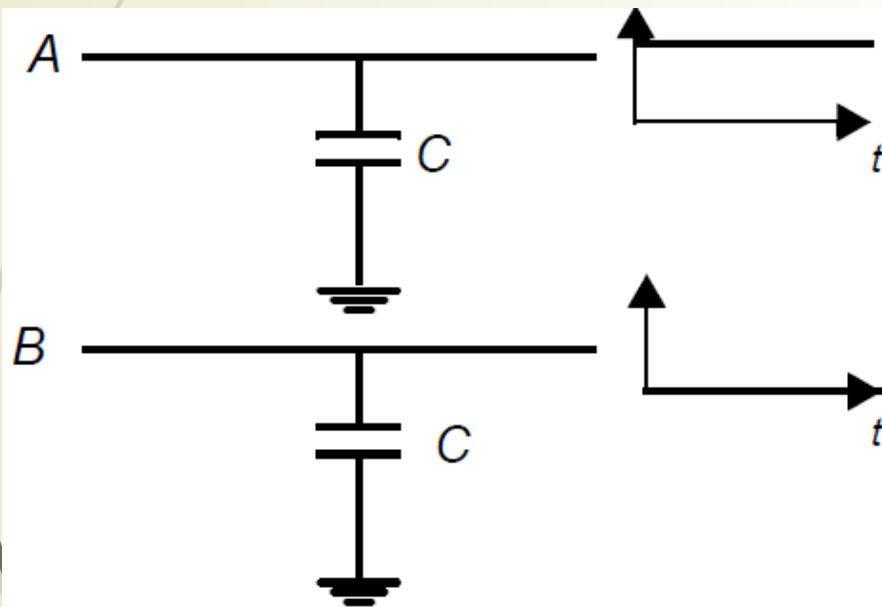
$$P_{(B=1)} = 0.2$$

$$P_{(C=1)} = 0.1$$

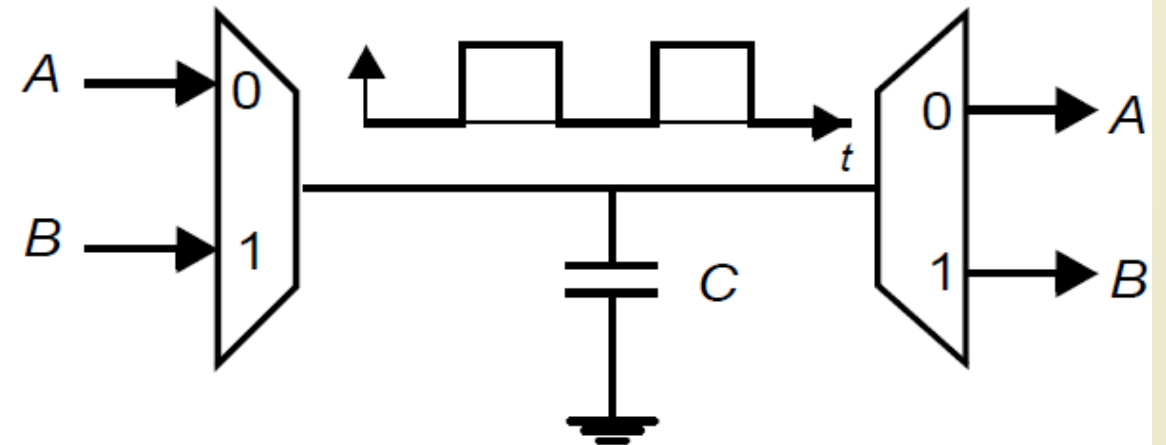
Figure Reordering of inputs affects the circuit activity.

Time-multiplexing resources

- Time-multiplexing a single hardware resource—such as a logic unit or a bus—over a number functions is an often used technique to minimize the implementation area. Unfortunately, the minimum area solution does not always result in the lowest switching activity.
- The degree of time-multiplexing should not affect the switched capacitance, since the time-multiplexed solution has half the capacitance switched.



(a) parallel data transmission



(b) serial data transmission

Figure Parallel versus time-multiplexed data busses.

Continue.....

- However if the data signals have some distinct properties (called temporal correlation), the power dissipation of the time-multiplexed solution can be significantly higher.
- Suppose, for instance, that A is always (or mostly) 1 and B is (mostly) 0. In the parallel solution, the switched capacitance is very low since there are very few transitions on the data bits. However, in the time-multiplexed solution, the bus toggles between 0 and 1.

Ratioed Logic

- ▶ Ratioed logic is used to reduce the number of transistors count but at the cost of extra power dissipation and reduced robustness.
- ▶ In ratioed logic, the entire PUN is replaced with a single unconditional load device.
- ▶ The nominal high voltage (V_{OH}) is V_{DD} but the nominal low voltage (V_{OL}) is not zero which results a static power dissipation.
- ▶ This also reduced the noise margin.
- ▶ Since, the output voltage depends on the sizes of the transistor so it is called ratioed logic.

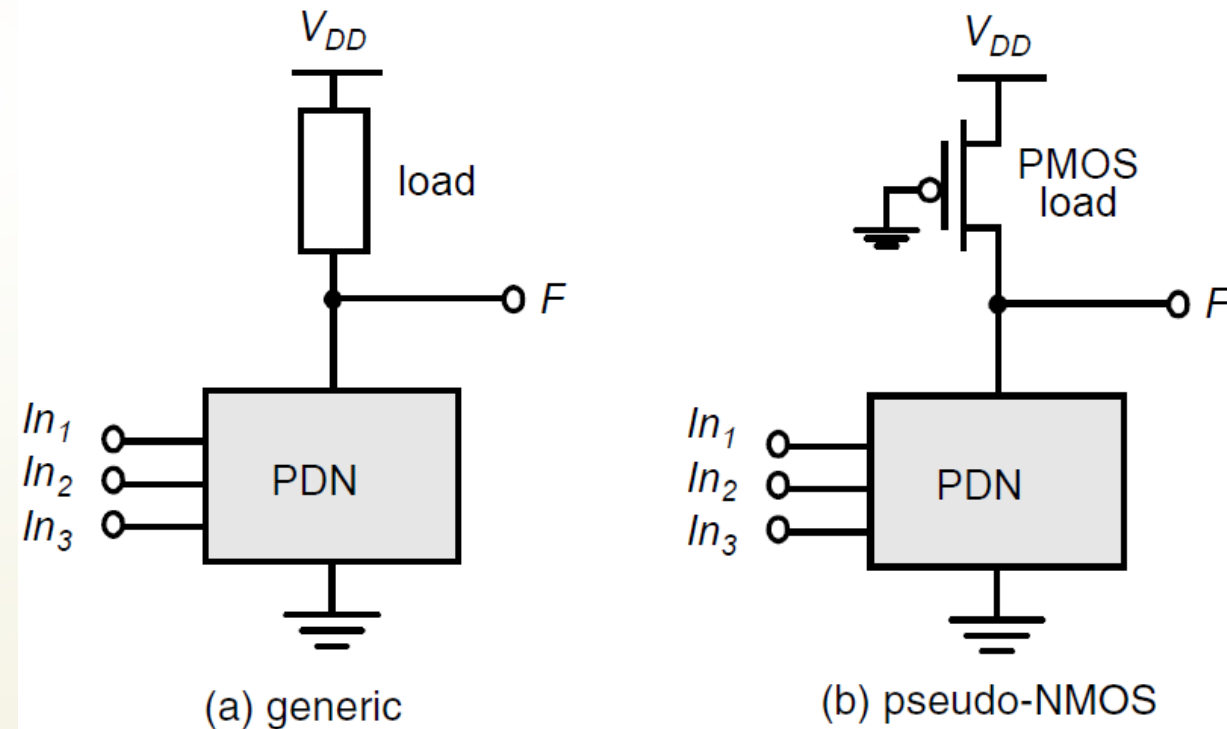


Figure Ratioed logic gate

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- The sizing of the load device relative to the pull-down devices can be used to trade-off parameters such a noise margin, propagation delay and power dissipation.
- Since the voltage swing on the output and the overall functionality of the gate depends upon the ratio between the NMOS and PMOS sizes, the circuit is called ratioed.

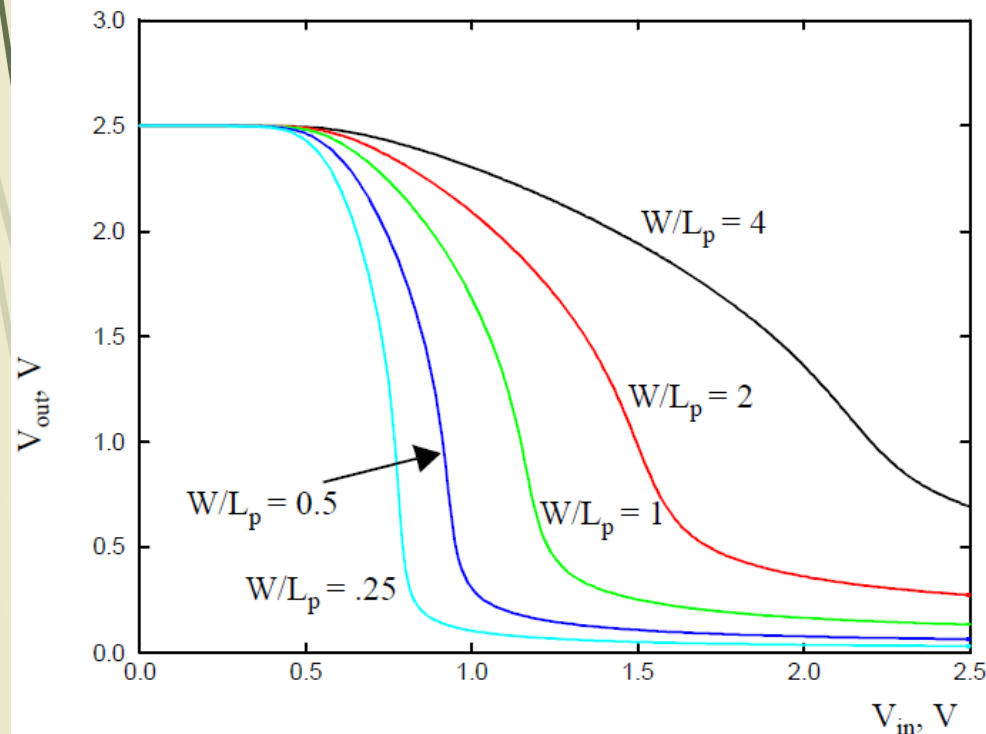


Figure Voltage-transfer curves of the pseudo-NMOS inverter as a function of the PMOS size.

$$V_{OL} \approx \frac{k_p(-V_{DD} - V_{Tp}) \cdot V_{DSAT}}{k_n(V_{DD} - V_{Tn})} \approx \frac{\mu_p \cdot W_p}{\mu_n \cdot W_n} \cdot |V_{DSAT}|$$

Table Performance of a pseudo-NMOS inverter.

Size	V_{OL}	Static Power Dissipation	t_{plh}
4	0.693V	564 μ W	14ps
2	0.273V	298 μ W	56ps
1	.133V	160 μ W	123ps
0.5	0.064V	80 μ W	268ps
0.25	0.031V	41 μ W	569ps

Differential Cascode Voltage Switch Logic (DCVSL)

- It is possible to create a ratioed logic style that completely eliminates static currents and provides rail-to-rail swing. Such a gate combines two concepts: differential logic and positive feedback.
- A differential gate requires that each input is provided in complementary format, and produces complementary outputs. An example of such a logic family, called Differential Cascode Voltage Switch Logic (or DCVSL) is presented in figure.
- when PDN1 conducts, PDN2 is off, and when PDN1 is off, PDN2 conducts.
- Assume now that, for a given set of inputs, PDN1 conducts while PDN2 does not, and that Out and $\overline{\text{Out}}$ are initially high and low, respectively.
- Turning on PDN1, causes Out to be pulled down, although there is still a fight between M1 and PDN1.

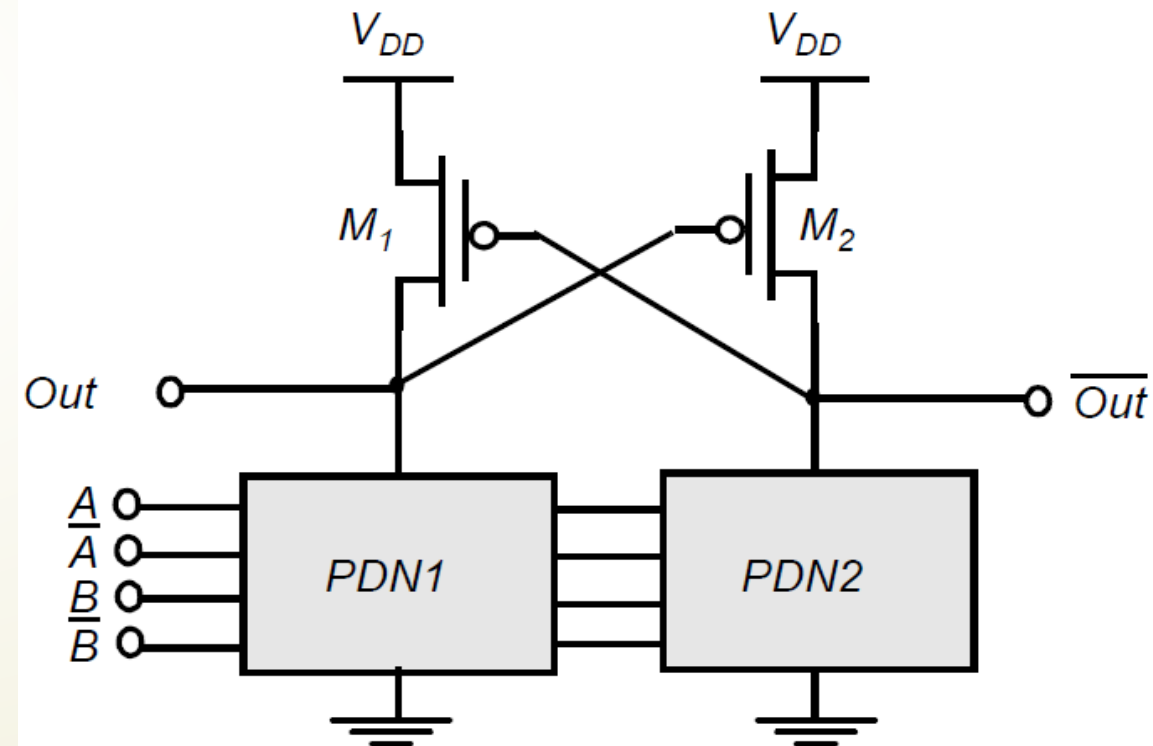
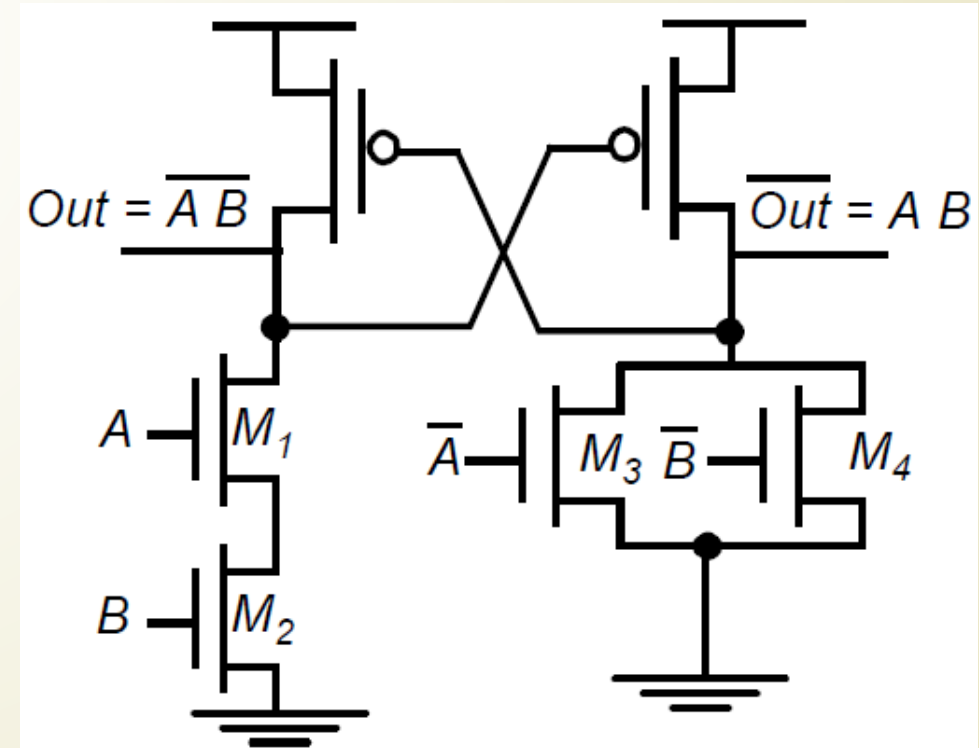


Figure Basic principle

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- At one point M2 turns on and starts charging Out to V_{DD} —eventually turning off M1. This in turn enables Out to discharge all the way to GND.
- The resulting circuit exhibits a rail-to-rail swing, and the static power dissipation is eliminated.
- The circuit is still ratioed since the sizing of the PMOS devices relative to the pull-down devices is critical. In addition to the problem of increase complexity in design. This circuit style still has a power-dissipation problem.



Design Consideration: Single-ended versus Differential

- ▶ A distinct advantage in DCVSL, it eliminates the need for an extra inverter to produce the complementary signal. It has been observed that a differential implementation of a complex function may reduce the number of gates required. For example in some logic design it often happens that both a signal and its complement are needed simultaneously.
- ▶ When the complementary signal is generated using an inverter, the inverted signal is delayed with respect to the original. This causes timing problems, especially in very high-speed designs. The differential output capability avoids this problem.

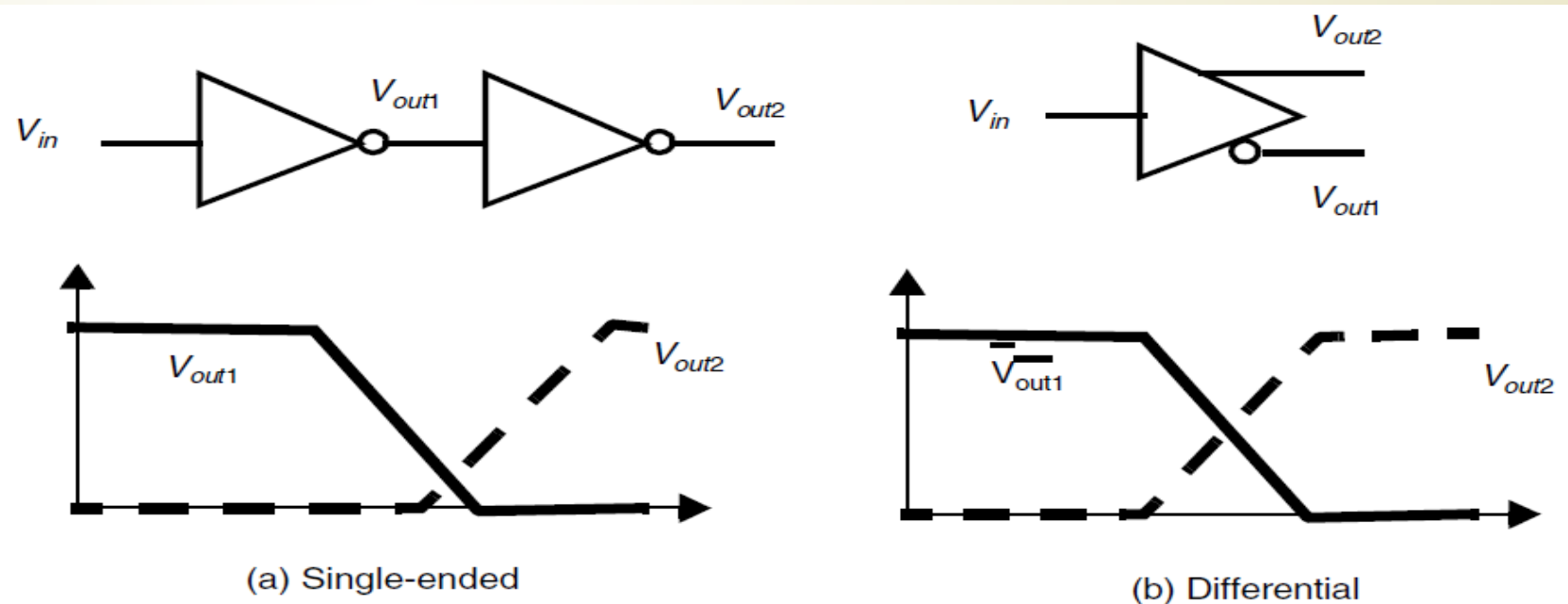


Figure Advantage of differential (b) over single-ended (a) gate

Continue....

- ▶ With all these positive properties, why not always use differential logic?
- ▶ Well, the differential nature virtually doubles the number of wires that has to be routed, leading very often to unwieldy designs (on top of the additional implementation overhead in the individual gates). Additionally, the dynamic power dissipation is high

Pass Transistors Basics

- ▶ A popular and widely-used alternative to complementary CMOS is pass-transistor logic, which attempts to reduce the number of transistors. The reduced number of devices has the additional advantage of lower capacitance.
- ▶ An implementation of the AND function constructed that way, using only NMOS transistors. In this gate, if the B input is high, the top transistor is turned on and copies the input A to the output F. When B is low, the bottom pass transistor is turned on and passes a 0.
- ▶ The switch driven by $\overline{B}$ seems to be redundant at first glance. Its presence is essential to ensure that the gate is static, this is that a low-impedance path exists to the supply rails under all circumstances.
- ▶ Consider the case when the pass transistor is charging up a node with the gate and drain terminals set at V_{DD} . Let the source of the NMOS pass transistor be labeled x. Node x will charge up to $V_{DD} - V_{Tn}(V_x)$.

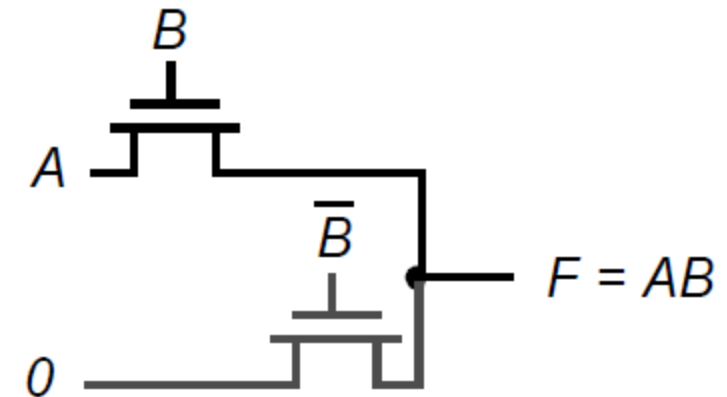


Figure Pass-transistor implementation of an AND gate

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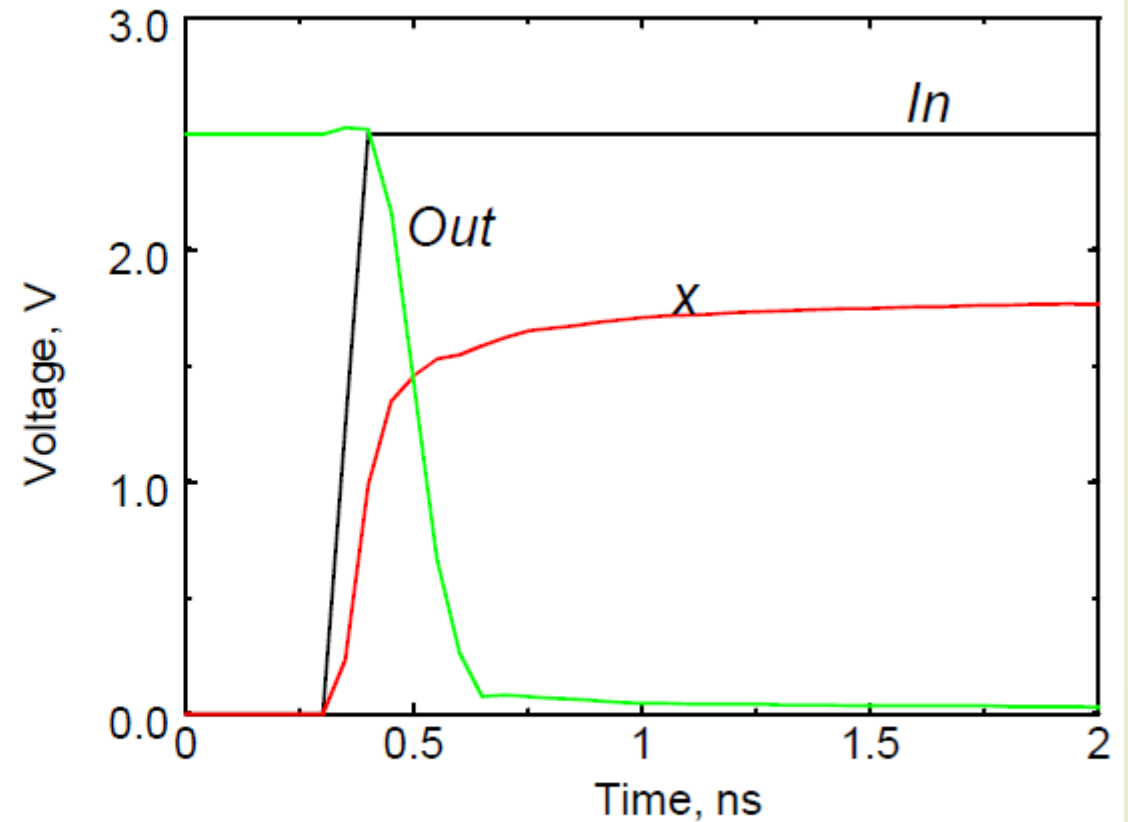
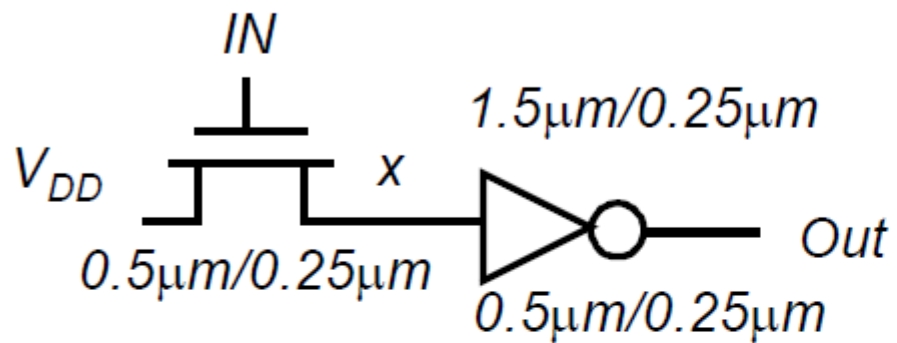
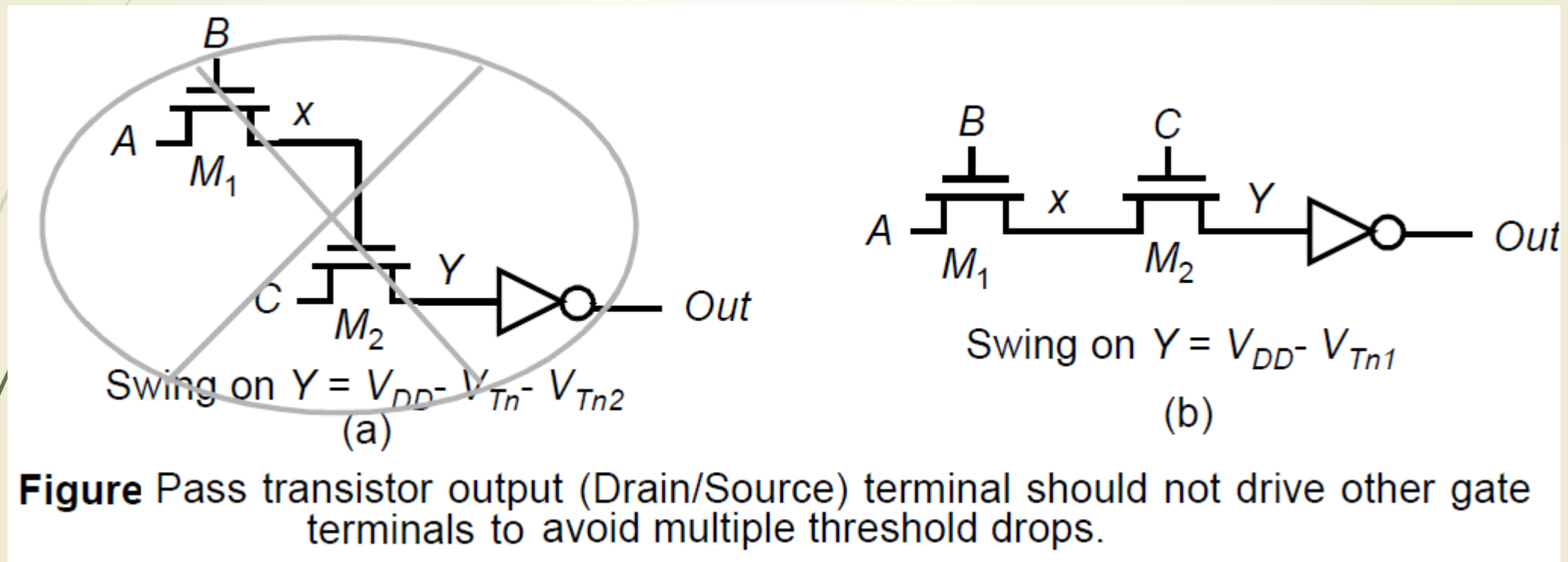


Figure Transient response of charging up a node using an N device. Notice the slow tail after an initial quick response.

Continue...

- ▶ The above example demonstrates that pass-transistor gates cannot be cascaded by connecting



VTC of the pass transistor AND gate

- Similar to complementary CMOS, the VTC of pass transistor logic is data-dependent.
- For the case when $B = V_{DD}$, the top pass transistor is turned on, while the bottom one is turned off. In this case, the output just follows the input A until the input is high enough to turn off the top pass transistor (i.e., reaches $V_{DD} - V_{Tn}$).
- Next consider the case when $A = V_{DD}$, and B makes a transition from 0 $\rightarrow$ 1. the bottom pass transistor is turned on till then and the output is close to zero.
- A similar behavior is observed when both inputs A and B transition from 0 $\rightarrow$ 1.

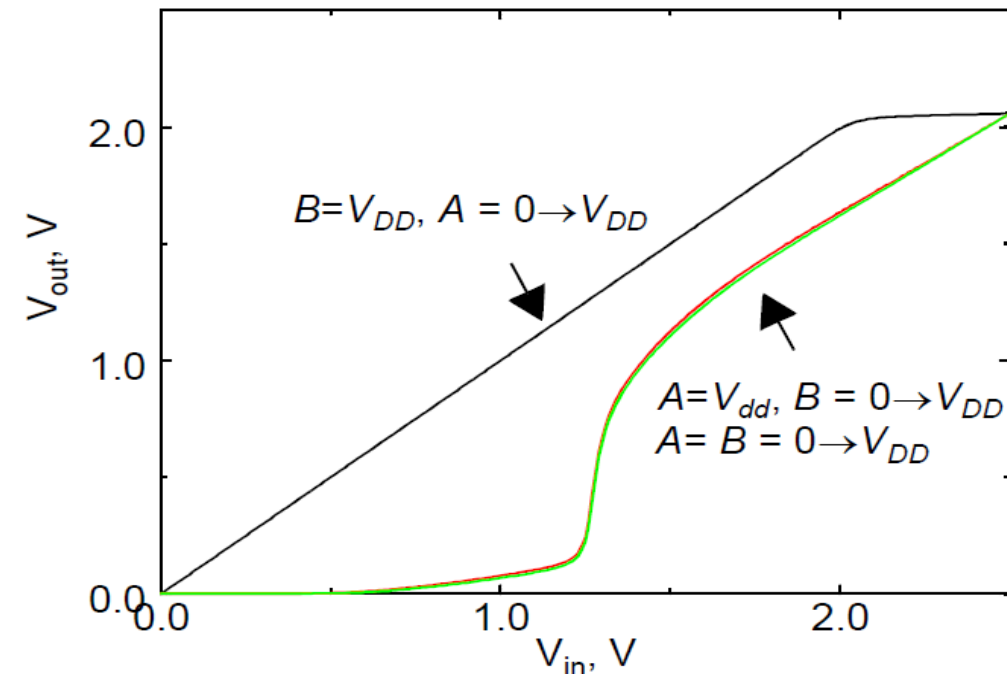
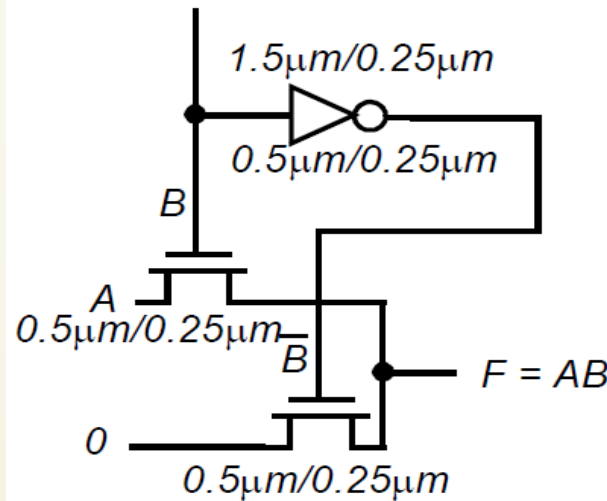


Figure Voltage Transfer Characteristic for the pass-transistor AND gate

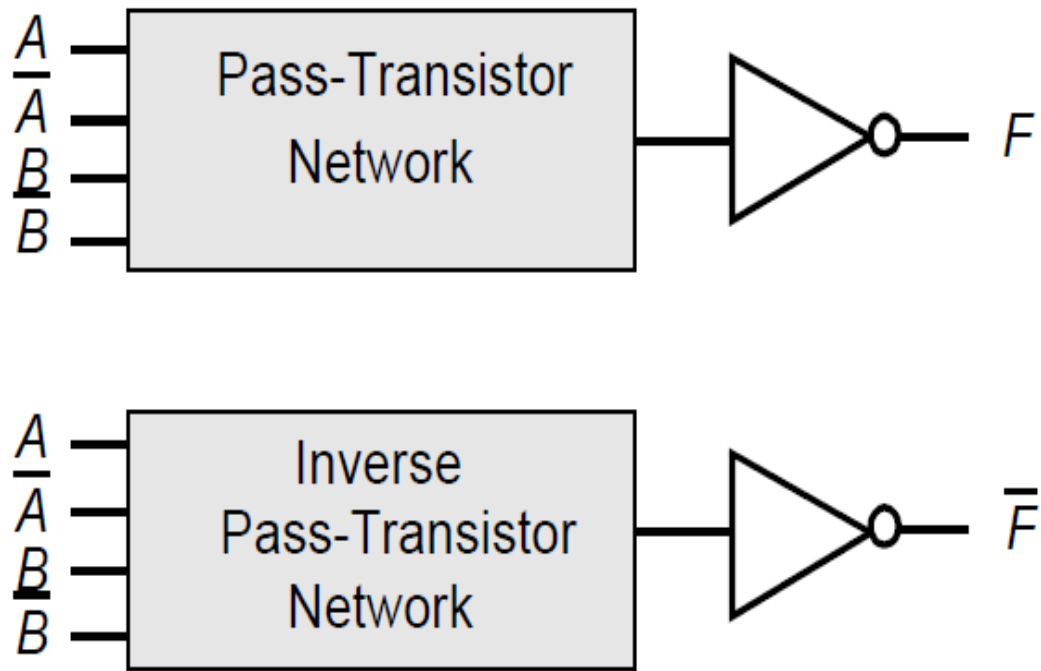
Continue.....

- ▶ Pass-transistors require lower switching energy to charge up a node due to the reduced voltage swing.
- ▶ For the pass transistor circuit in previous case assume that the drain voltage is at V_{DD} and the gate voltage transitions to V_{DD} .
- ▶ The output node charges from 0V to $V_{DD}-V_{Tn}$ (assuming that node x was initially at 0V). and the energy drawn from the power supply for charging the output of a pass transistor is given by:

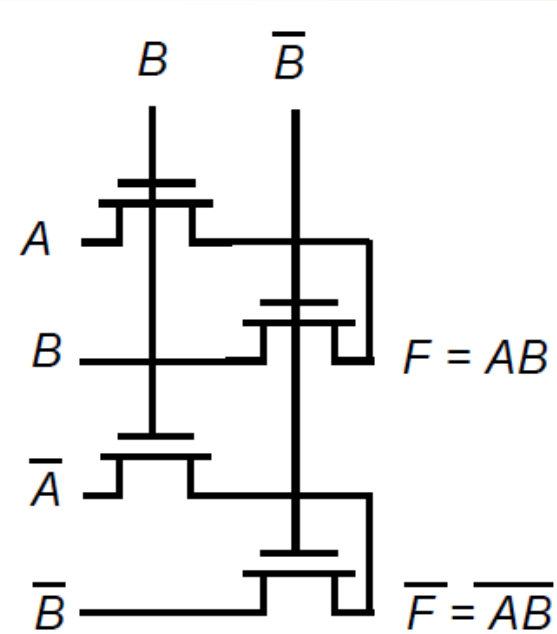
$$E_{0 \rightarrow 1} = \int_0^T P(t) dt = V_{DD} \int_0^T i_{supply}(t) dt = V_{DD} \int_0^{(V_{DD}-V_{Tn})} C_L dV_{out} = C_L \cdot V_{DD} \cdot (V_{DD}-V_{Tn})$$

Differential Pass Transistor Logic

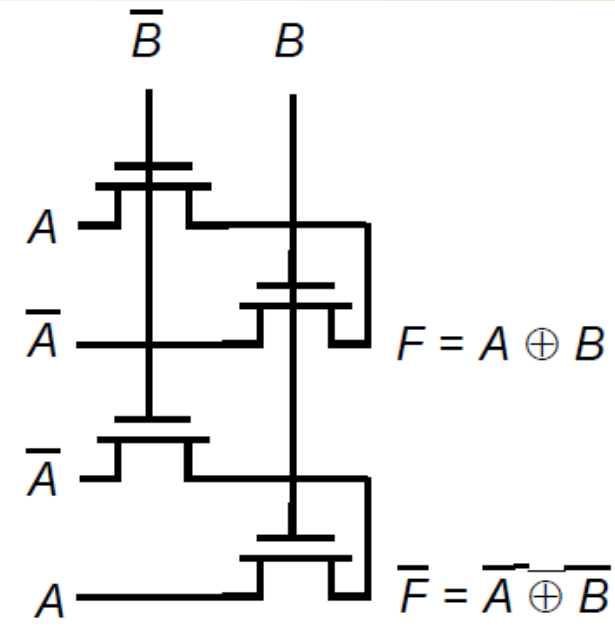
- ▶ A differential pass-transistor logic family, called CPL or DPL, is commonly used. The basic idea (similar to DCVSL) is to accept true and complementary inputs and produce true and complementary outputs.
- ▶ A number of CPL gates (AND/NAND and XOR/NXOR) are shown in Figure



(a) Basic concept



AND/NAND



XOR/NXOR

Figure Complementary pass-transistor logic (CPL).

Continue.....

- ▶ These gates possess a number of interesting properties -
- ▶ Some complex gates such as XORs and adders can be realized efficiently with a small number of transistors. Furthermore, the availability of both polarities of every signal eliminates the need for extra inverters.
- ▶ CPL belongs to the class of static gates, because the output-defining nodes are always connected to either VDD or GND through a low resistance path. This is advantageous for the noise resilience.
- ▶ The design is very modular. In effect, all gates use exactly the same topology. Only the inputs are permuted. This makes the design of a library of gates very simple.

Robust and Efficient Pass-Transistor Design

- Differential pass-transistor logic, like single-ended pass-transistor logic, suffers from static power dissipation and reduced noise margins. since the high input to the signal-restoring inverter only charges up to $V_{DD}-V_{Tn}$.
- There are several solutions proposed to deal with this problem as outlined below.

Solution 1: Level Restoration

- A common solution to the voltage drop problem is the use of a level restorer, which is a single PMOS configured in a feedback path.
- If input A makes a 0 to V_{DD} transition, Mn only charges up node X to $V_{DD}-V_{Tn}$. This is, however, enough to switch the output of the inverter low, turning on the feedback device Mr and pulling node X all the way to V_{DD} .
- This eliminates any static power dissipation in the inverter because voltage levels are either at GND or VDD, and no static power is consumed. This solution add one complexity since the circuit is now ratioed

Continue...

- ▶ If W/L ratio is large it will transfer the charge at faster rate. Furthermore it will store more value.

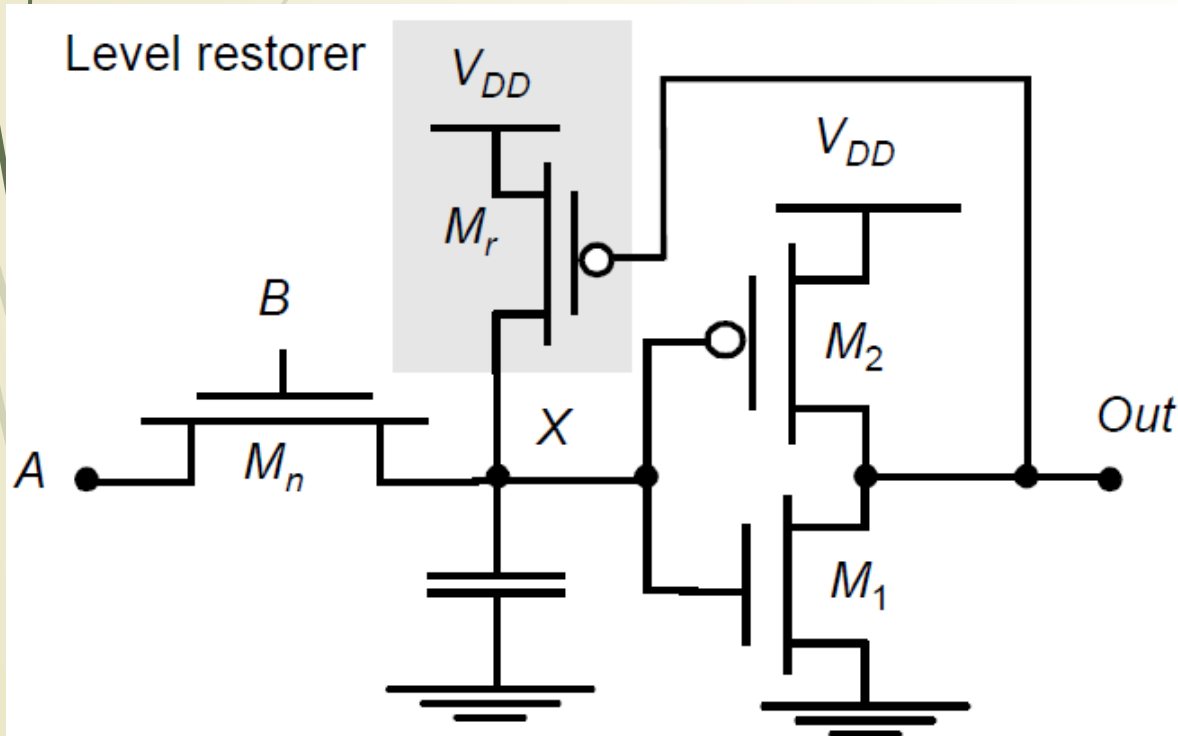


Figure Level-restoring circuit.

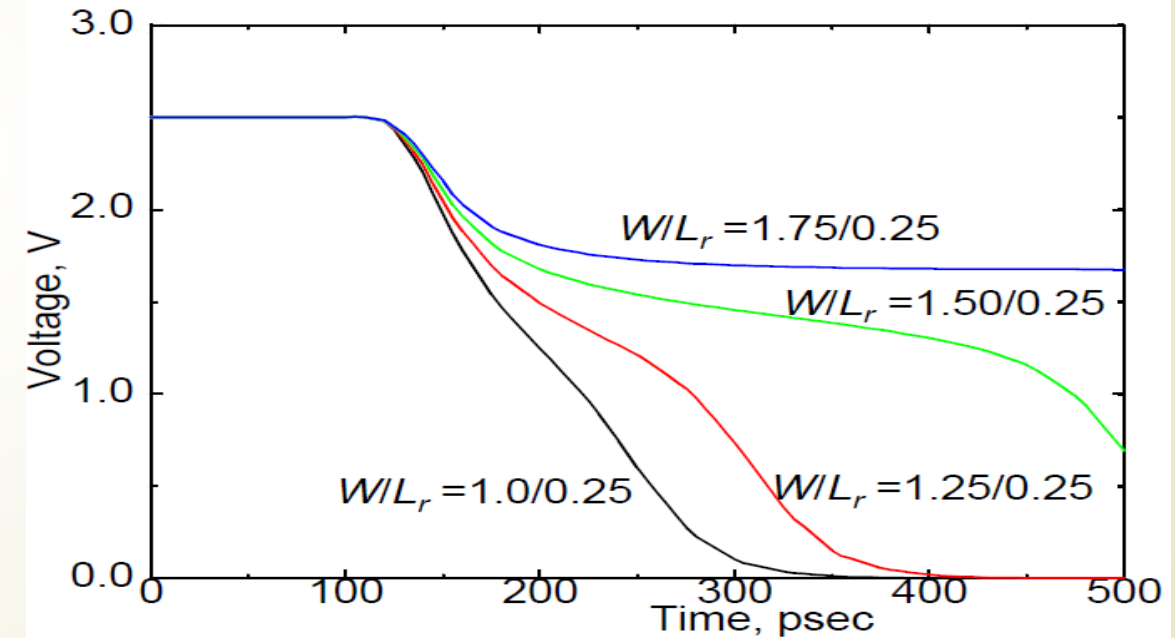


Figure Transient response of the circuit in Figure 6.40. A level restorer that is too large can result in incorrect evaluation.

Continue...

- ▶ We have consistently widened the PMOS transistor so that its resistance matches. The motivation behind this approach is to create an inverter with symmetrical VTC. However, this does not imply that this ratio also yields the minimum overall propagation delay.
- ▶ If symmetry and noise margins are not of prime concern it is actually possible to speed up the inverter by reducing the width of the PMOS device.
- ▶ This optimum ratio can be derived through the following simple analysis. Consider two identical, cascaded CMOS inverters. The load capacitance of the first gate equals approximately

Sizing of level restorer

- ▶ The sizing of M_r and M_n will change the logic at point 'x'.
- ▶ If M_r is stronger than M_n restoring at x is faster.
- ▶ If M_n is stronger than M_r de-storing at x is an important issue.
- ▶ Power dissipation is very small – no static power dissipation.
- ▶ Another concern if delay of inverter is more then M_r transistor will take more time to restore.

Swing-restored pass transistor logic

- A modification of the level-restorer, applicable in differential networks and known as swing-restored pass transistor logic. Instead of a simple inverter or half-latch at the output of the pass transistor network, two back-to-back inverters, configured in a cross-coupled fashion, are used for level restoration and performance improvement.

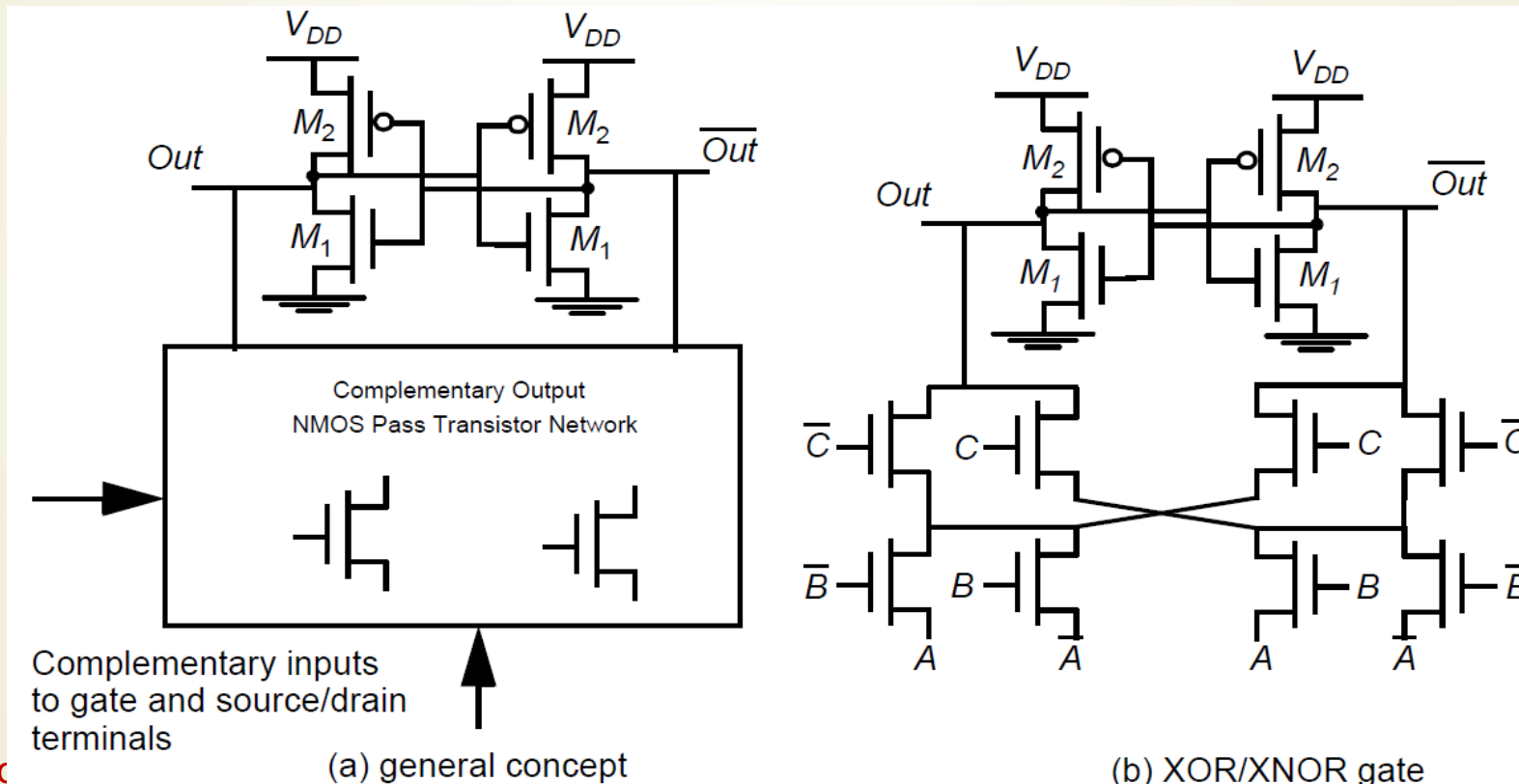


Figure Swing-restored pass transistor logic

Multiple-Threshold Transistors

- ▶ If we use zero threshold devices for the NMOS pass-transistors eliminates most of the threshold drop, and passes a signal close to V_{DD} .
- ▶ The use of multiple-threshold transistors is becoming more common, and involves simple modifications to existing process flows.
- ▶ The use of zero-threshold transistors can be dangerous
- ▶ due to the subthreshold currents that can flow through
- ▶ the pass-transistors, even if V_{GS} is slightly below V_T .

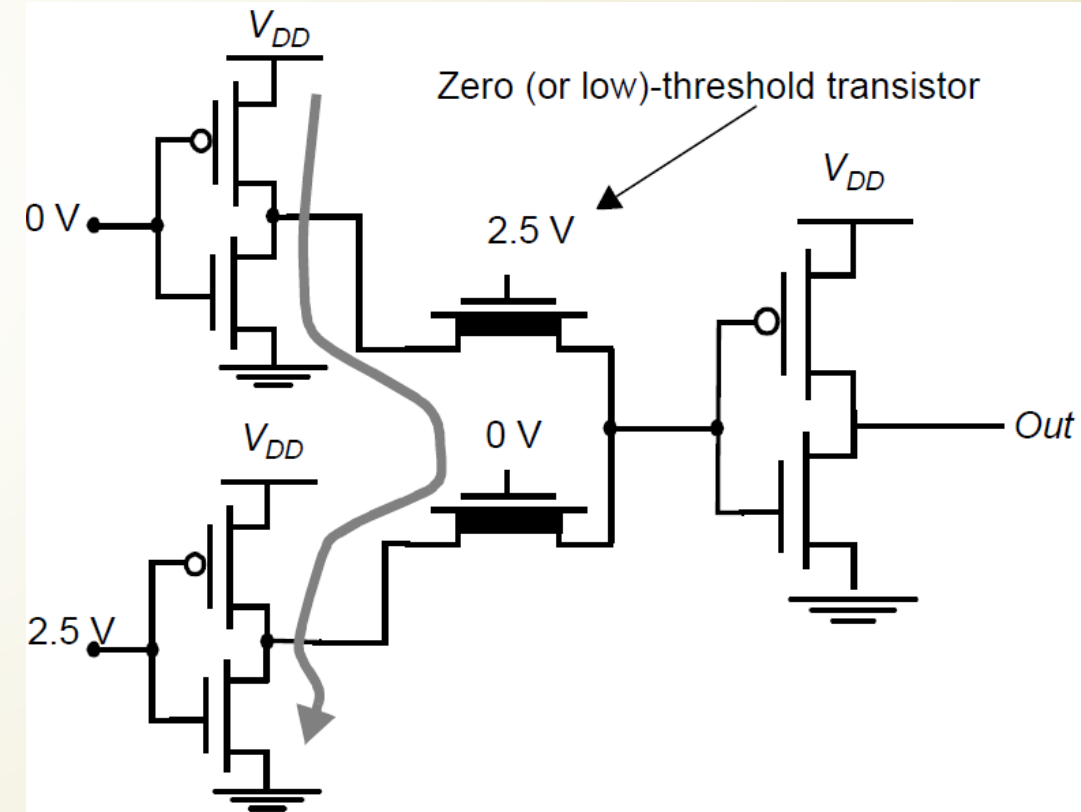


Figure Static power consumption when using zero-threshold pass-transistors.

Transmission Gate Logic

- ▶ The most widely-used solution to deal with the voltage-drop problem is the use of transmission gates.
- ▶ The transmission gate combines the best of both device flavors by placing a NMOS device in parallel with a PMOS device.
- ▶ The transmission gate acts as a bidirectional switch controlled by the gate signal C. In short

$$A = B \quad \text{if} \quad C = 1$$

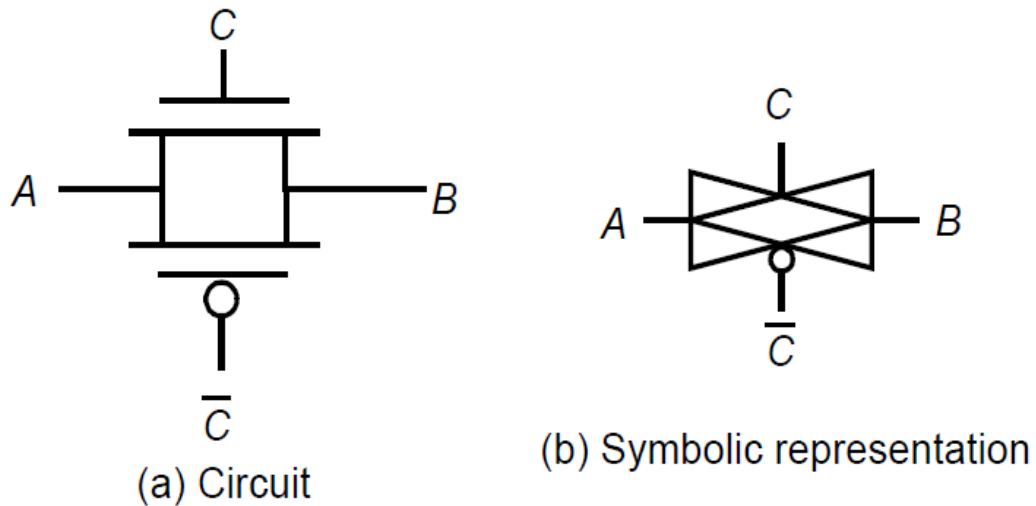


Figure CMOS transmission gate

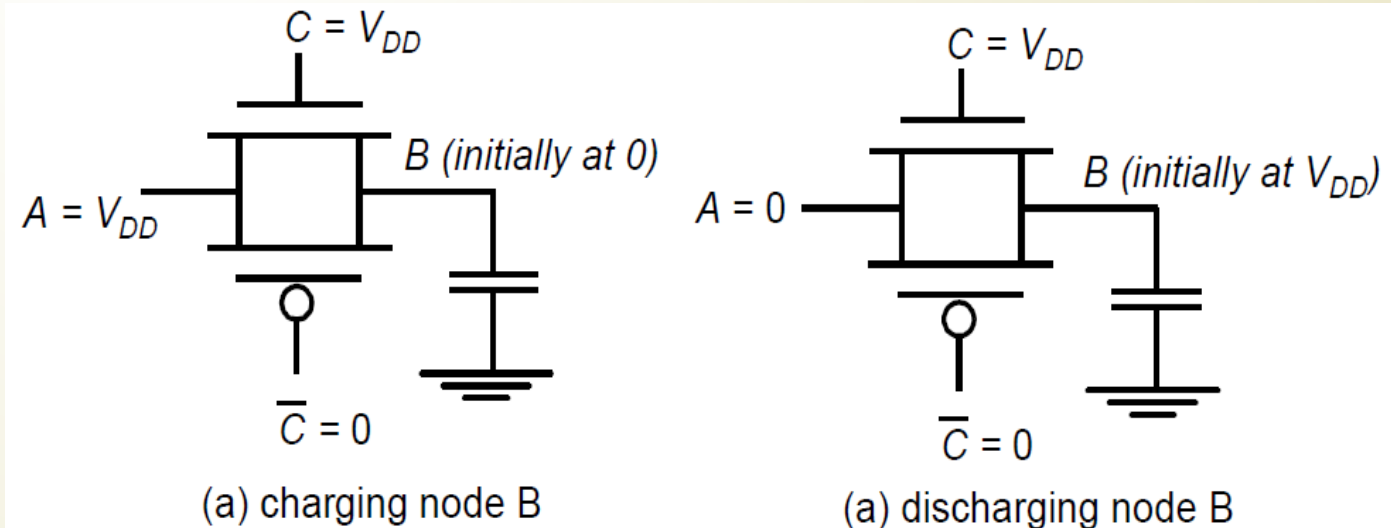


Figure Transmission gates enable rail-to-rail switching

Continue.....

- Transmission gates can be used to build some complex gates very efficiently . The gate either selects input A or B based on the value of the control signal S, which is equivalent to implementing the following Boolean function.

$$\bar{F} = (A \cdot S + B \cdot \bar{S})$$

- A complementary implementation of the gate requires eight
- transistors instead of six

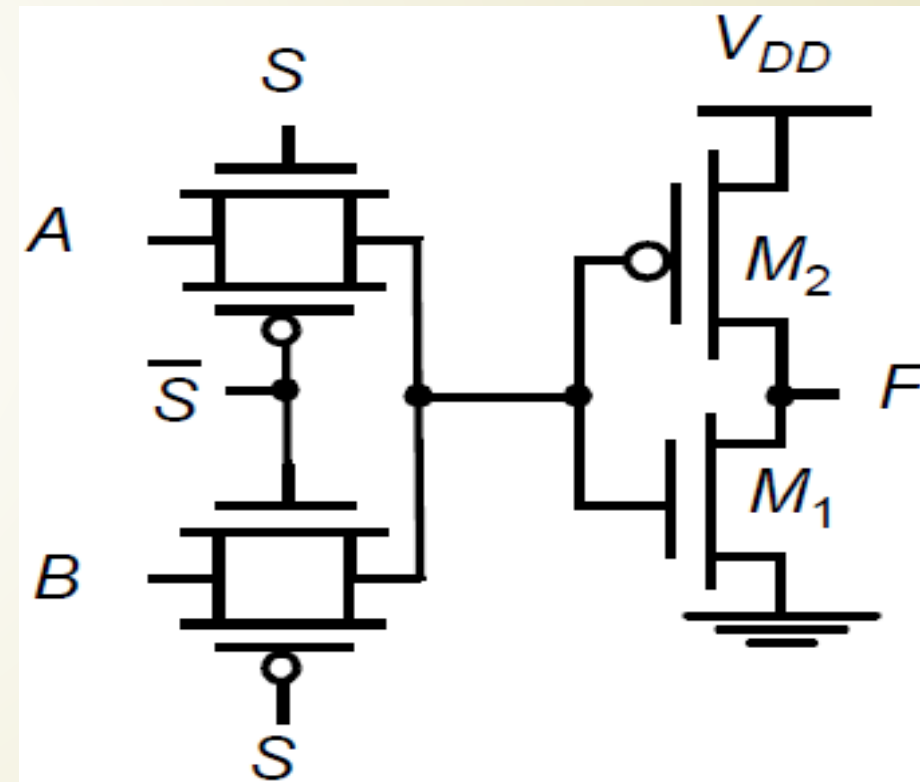


Figure Transmission gate multiplexer and its layout.

Continue.....

- ▶ Another example of the effective use of transmission gates is the popular XOR circuit.
- ▶ To understand the operation of this circuit, we have to analyze the $B = 0$ and $B = 1$ cases separately.
- ▶ For $B = 1$, transistors M_1 and M_2 act as an inverter while the transmission gate M_3/M_4 is off; hence $F = \overline{AB}$.
- ▶ In the opposite case, M_1 and M_2 are disabled, and the transmission gate is operational, or $F = A\overline{B}$. The combination of both results in the XOR function.

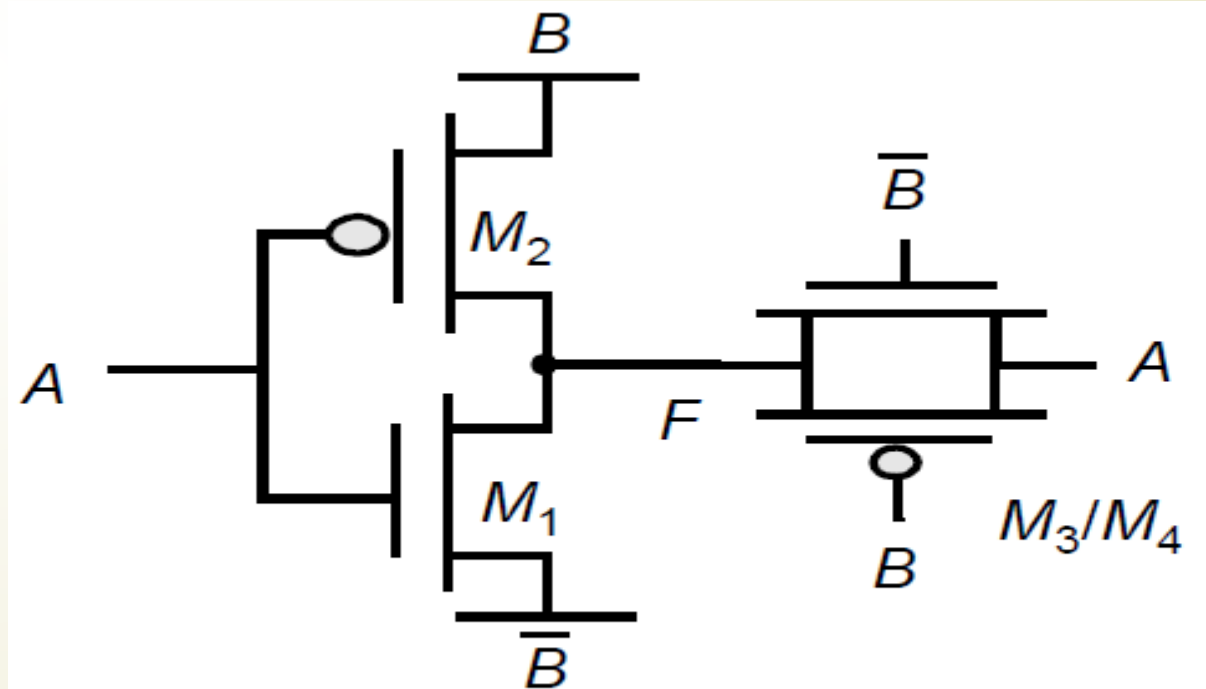


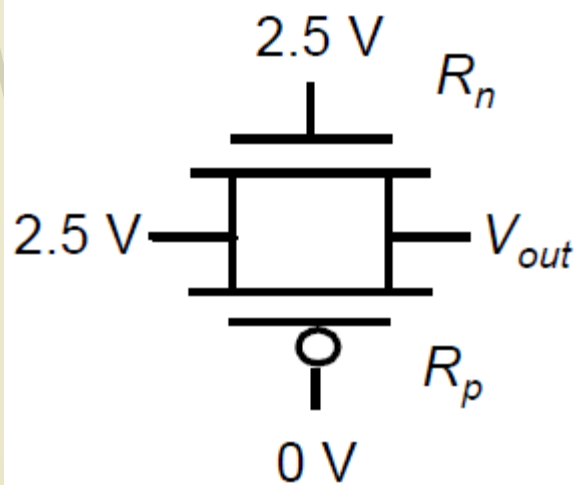
Figure Transmission gate XOR.

Performance of Pass-Transistor and Transmission Gate Logic

- ▶ The pass-transistor and the transmission gate are, unfortunately, not ideal switches, and have a series resistance associated with it.
- ▶ The effective resistance is modeled as a parallel connection of R_n and R_p .

$$R_p = \frac{V_{DD} - V_{out}}{I_p} = \frac{V_{DD} - V_{out}}{k_p \cdot \left((-V_{DD} - V_{Tp})(V_{out} - V_{DD}) - \frac{(V_{out} - V_{DD})^2}{2} \right)}$$

$$\approx \frac{1}{k_p (V_{DD} - |V_{Tp}|)}$$



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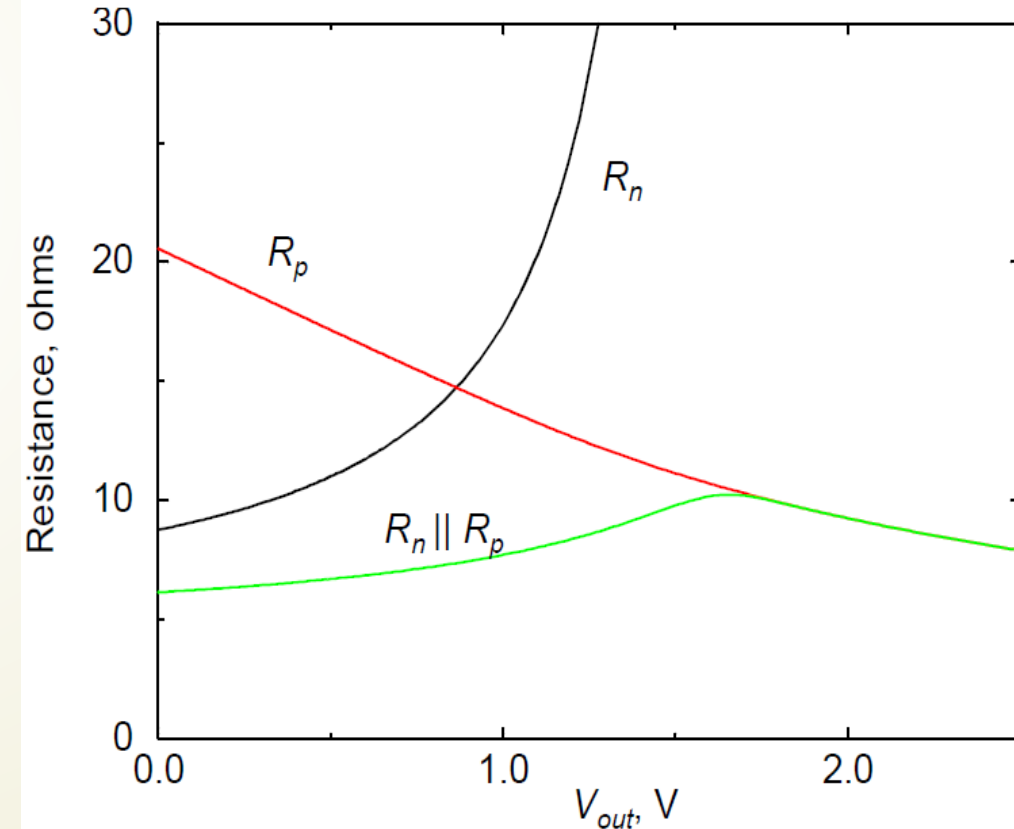


Figure Simulated equivalent resistance of transmission gate



Thank You

Rohit Lorenzo, VIT-AP

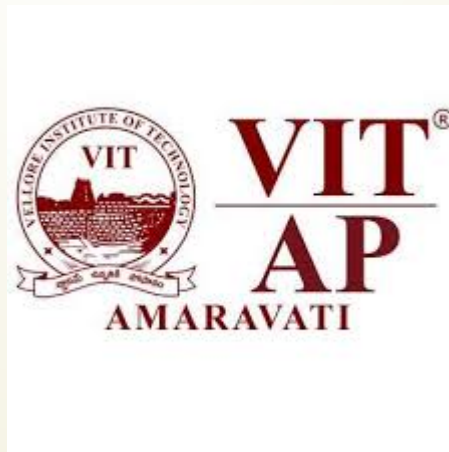
CMOS VLSI Design

By

Dr.Rohit Lorenzo

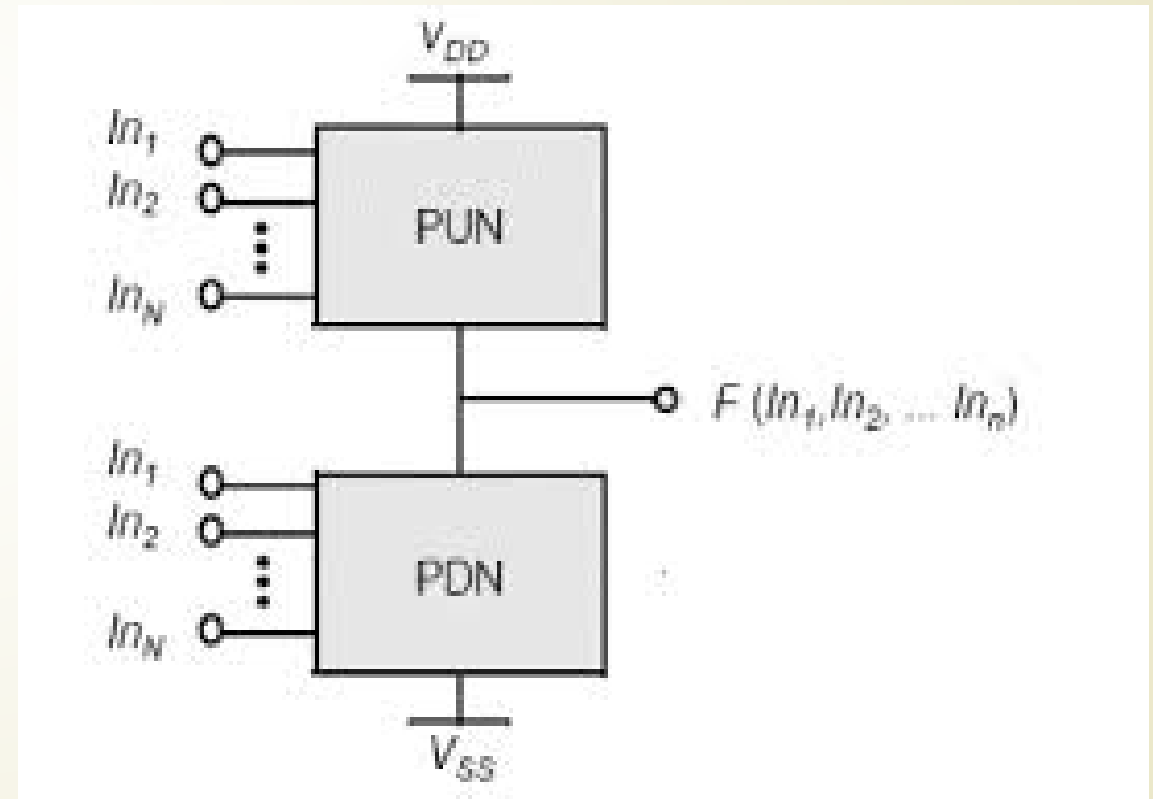
Assistant Professor Sr. Grade -1

School of Electronics Engineering (SENSE)



Static CMOS Complementary Logic

- Static CMOS gates are implemented by using combination of two networks, the pull-up network (PUN) and Pull-down Network (PDN).
- Advantage –
 - Very good current driving capability.
 - High Noise Margin.
 - Lower static power dissipation
- Dis-advantage –
 - Larger number of transistors required (larger chip area & delay).
 - Short circuit power dissipation.



Pseudo-nMOS Logic

- ▶ An n -input pseudo nMOS requires $n+1$ transistors compared to $2n$ transistors of the corresponding static CMOS gates.
- ▶ There is substantial reduction in area and delay.
- ▶ But, the static power dissipation is much higher.

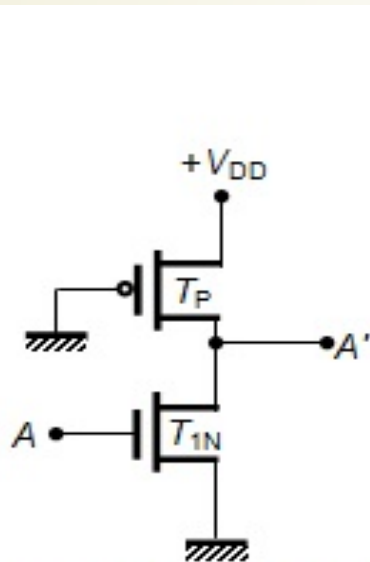


Fig. p-NMOS NOT

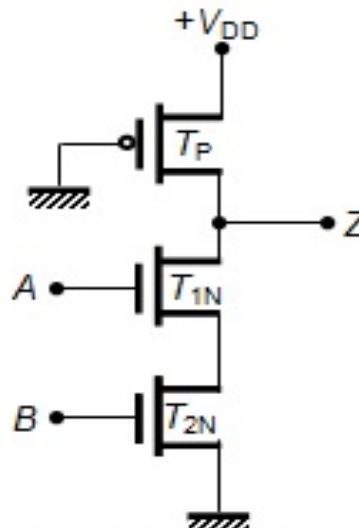


Fig. p-NMOS NAND

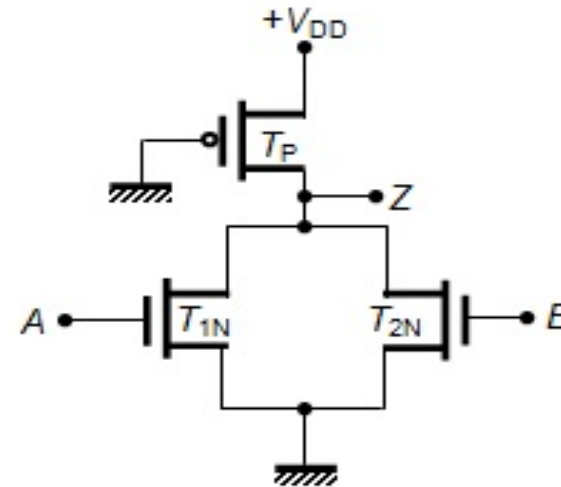


Fig. p-NMOS NOR

comparison of Static CMOS and Pseudo nMOS

	Static CMOS	Pseudo nMOS
1	Large area	Smaller area
2	Longer delay	Smaller delay
3	No Static power dissipation	Static power dissipation

Dynamic CMOS Design

- ▶ The basic construction of an (n-type) dynamic logic gate is shown in Figure. The PDN is constructed exactly as in complementary CMOS.
- ▶ The operation of this circuit is divided into two major phases: precharge and evaluation, with the mode of operation determined by the clock signal CLK.

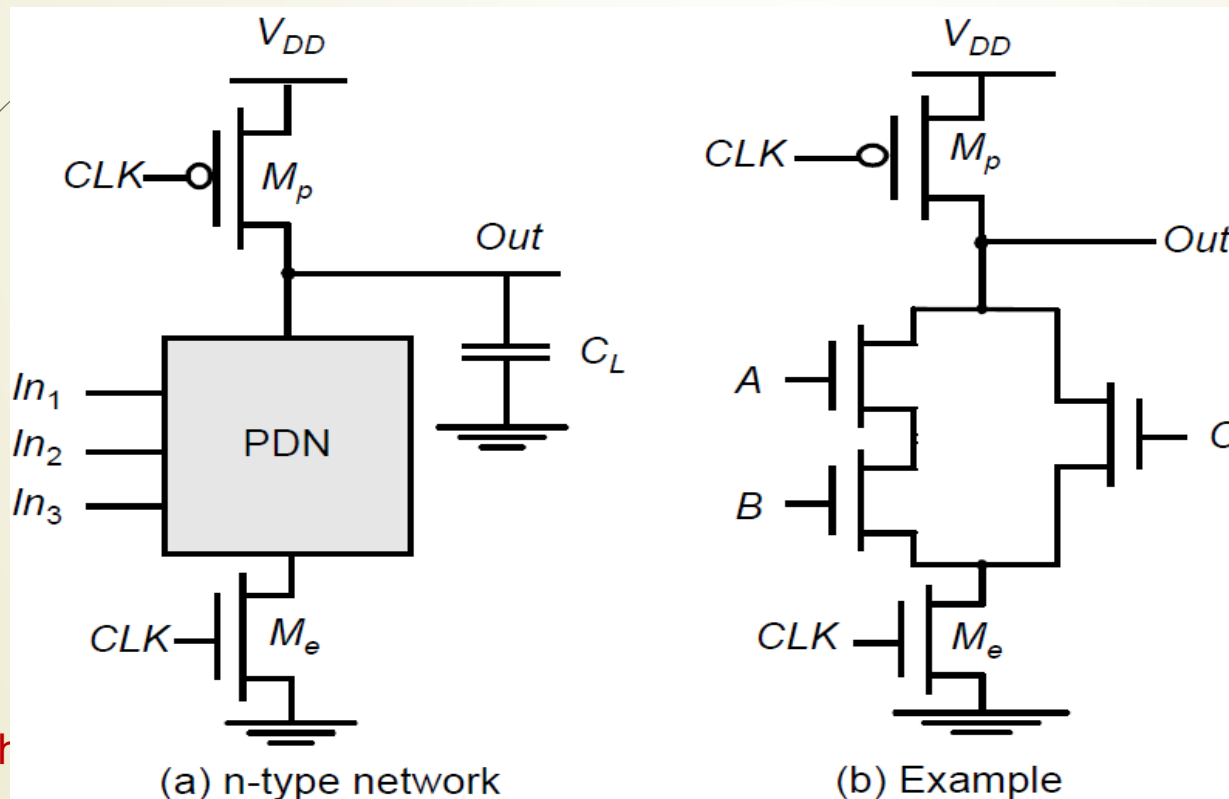


Figure Basic concepts of a dynamic gate

Continue.....

➤ Pre-charge Phase

- When **CLK = 0**, the output node Out is precharged to V_{DD} by the PMOS transistor **M_p**. During that time, the evaluate NMOS transistor **M_e** is off, so that the pull-down path is disabled.

➤ Evaluation Phase

- For **CLK = 1**, the precharge transistor **M_p** is off, and the evaluation transistor **M_e** is turned on. The output is conditionally discharged based on the input values and the pull-down topology.
- If the **PDN** is turned off, the precharged value remains stored on the output capacitance **C_L**. once Out is discharged, it cannot be charged again till then next precharge operation. Notice that the output can be in the high-impedance state during the evaluation period if the pull-down network is turned off.
- During the precharge phase (**CLK=0**), the output is precharged to V_{DD} regardless of the input values since the evaluation device is turned off. During evaluation (CLK=1), a conducting path is created between Out and GND if (and only if) $A \cdot B + C$ is TRUE. Otherwise, the output remains at the precharged state of V_{DD} .
- The following function is thus realized:

$$Out = \overline{CLK} + \overline{(A \cdot B + C)} \cdot CLK$$

Speed and Power Dissipation of Dynamic Logic

- ▶ The main advantages of dynamic logic are increased speed and reduced implementation area. Fewer devices to implement a given logic function implies that the overall load capacitance is much smaller.

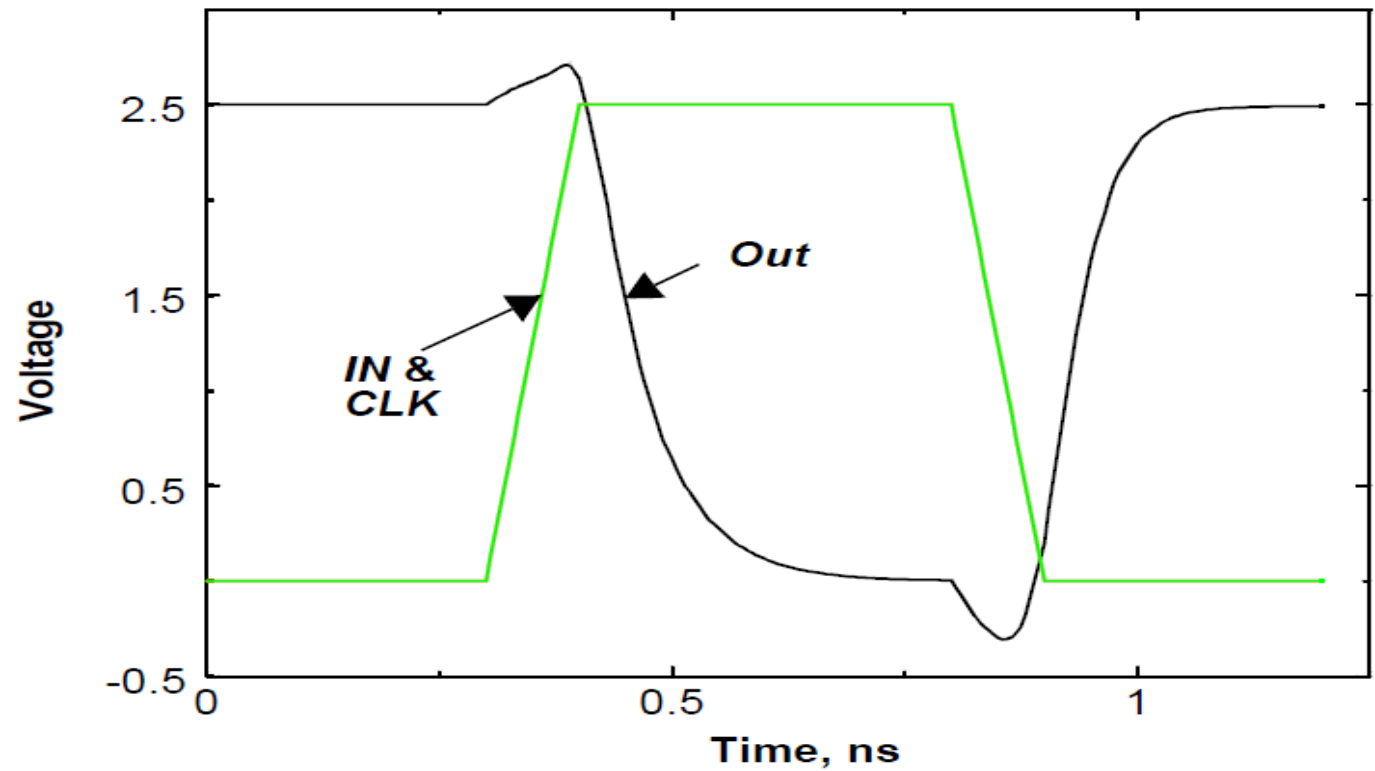
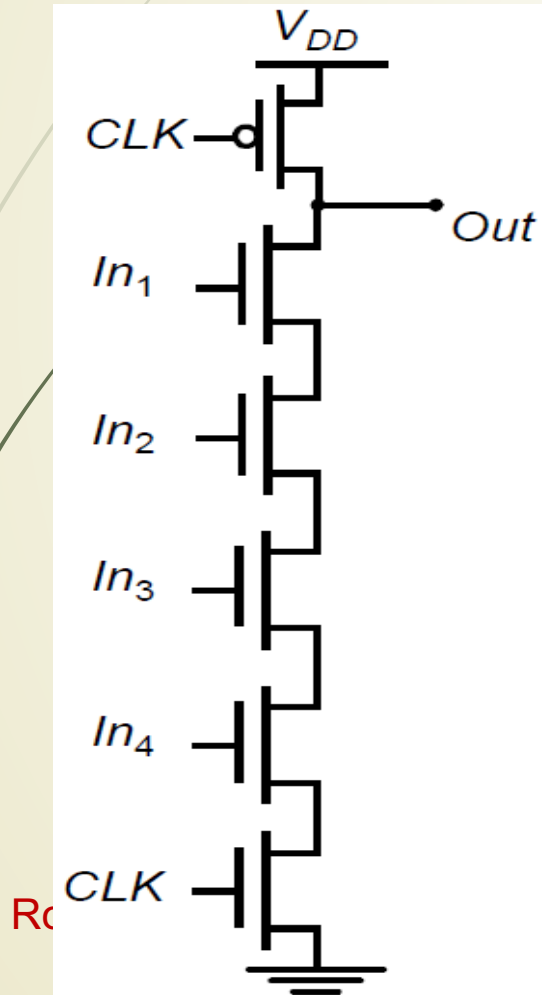
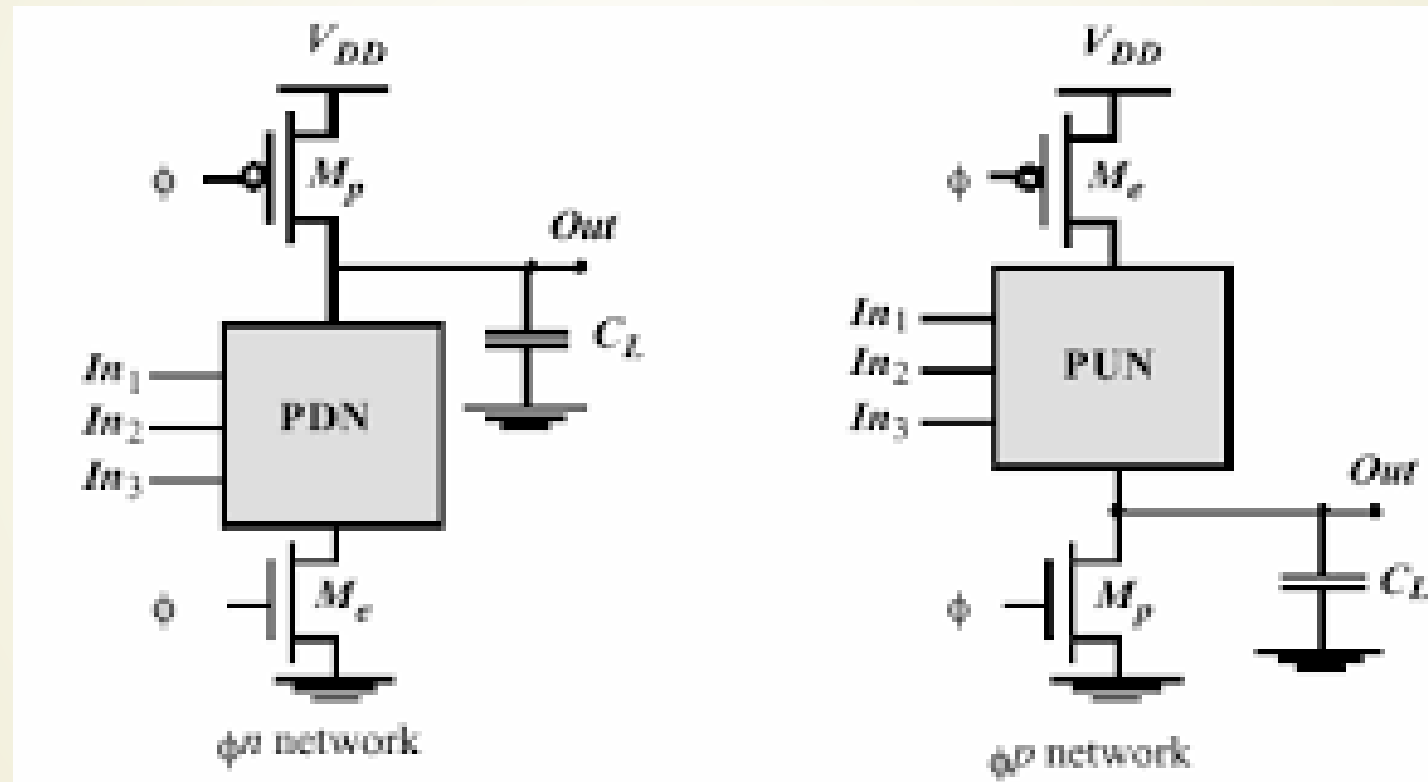


Figure Schematic and transient response of a four-input dynamic NAND gate.

Dynamic CMOS with Pull-up network(PUN) and Pull-down Network (PDN)



Advantages

- ▶ The number of transistor required for a circuit with fan-in N is $(N+2)$, in contract to $2N$ in case of static CMOS circuits.
- ▶ The load capacitance is more than 50% less than static CMOS and is closer to that of nMOS circuits.
- ▶ The speed of operation is faster than that of static CMOS circuits.
- ▶ Lower static power dissipation. (Its only consumes dynamic power)
- ▶ No short circuit power dissipation.
- ▶ No glitching power dissipation.
- ▶ It is non-ratioed. The sizing of the PMOS precharge device is not important for realizing proper functionality of the gate. The size of the precharge device can be made large to improve the low-to-high transition time. There is however, a trade-off with power dissipation since a larger precharge device directly increases clock-power dissipation.

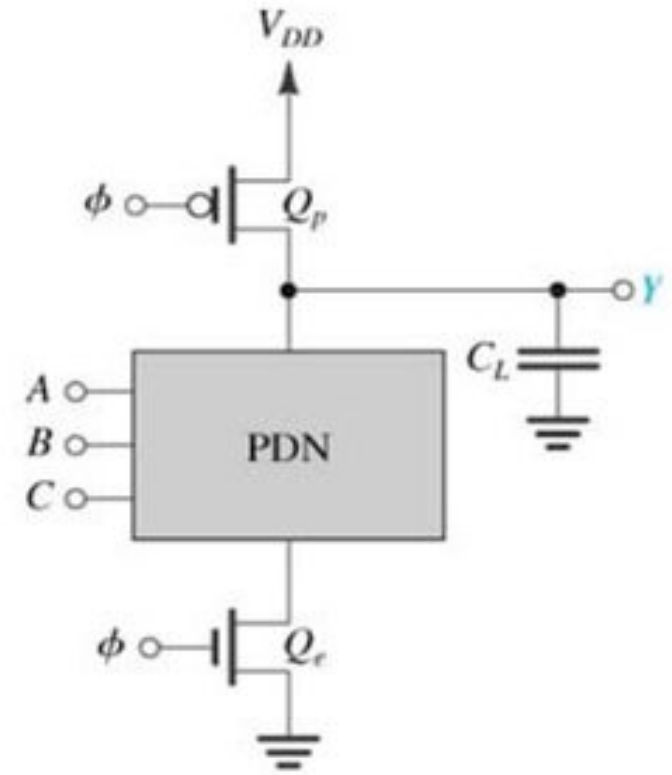


Disadvantages

- ▶ Charge leakage problem
- ▶ Charge sharing problem
- ▶ Clock Skew Problem

Charge Leakage Problem

- ▶ The operation of the dynamic gate depends on the storage of information in the form of charge on the capacitor.
- ▶ During pre-charge phase C_L is charge but how long the voltage on C_L is retain.
- Suppose clock period is too long, in that case capacitor is discharge through NMOS network.



Continue....

- ▶ The output should ideally remain at the precharged state of V_{DD} during the evaluation phase. However, this charge gradually leaks away due to leakage currents, eventually resulting in a malfunctioning of the gate

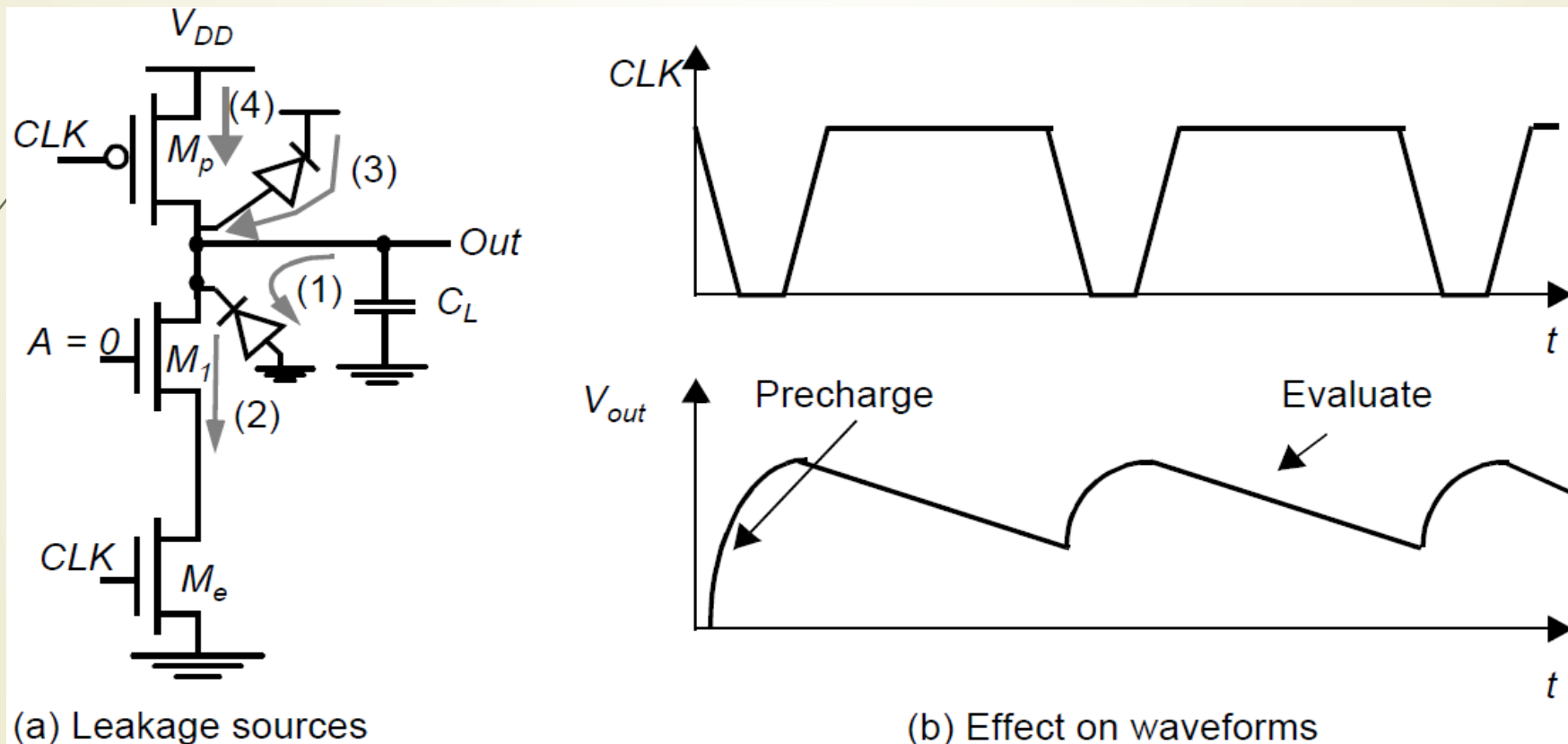
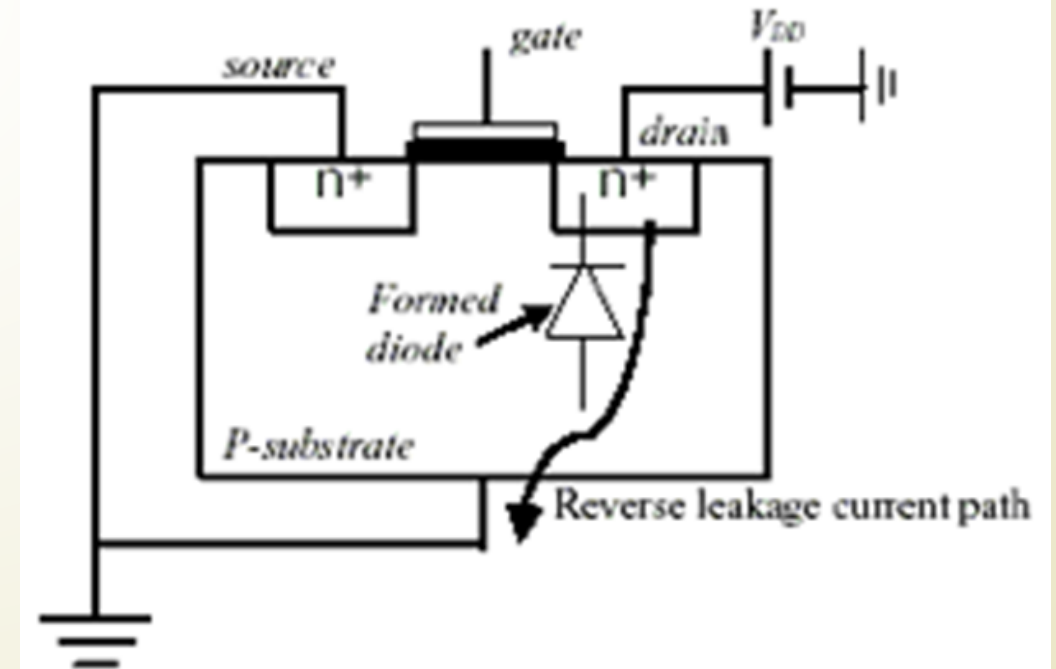


Figure Leakage issues in dynamic circuits.

Continue.....

- Source 1 and 2 are the reverse-biased diode and sub-threshold leakage of the NMOS pull-down device M1, respectively. The charge stored on C_L will slowly leak away due these leakage sources.
- Note that the PMOS precharge device also contributes some leakage current due to the reverse bias diode (source 3) and the subthreshold conduction (source 4).
- If this reverse current for long duration the capacitor will discharge leading to degradation in output level.
- The output may degrade to such a level, which will be treated as low level. So we need to prevent this.

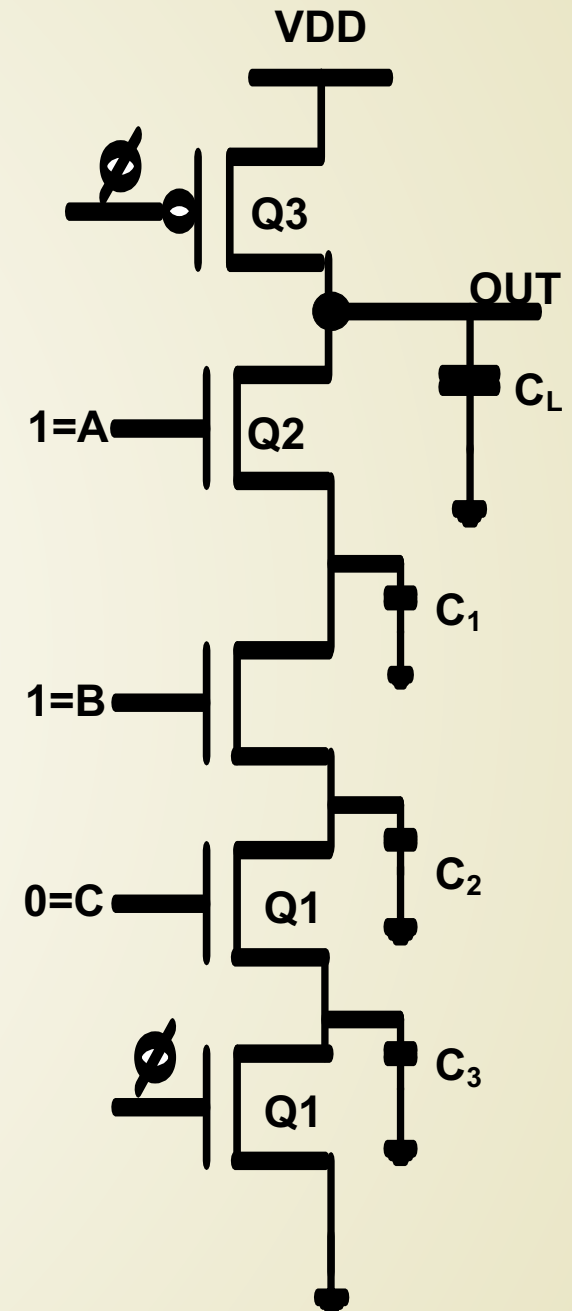


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- ▶ The leakage current is expressed by the diode equation – $I_D = I_S (e^{qV/KT} - 1)$
- ▶ Here is the reverse saturation current, V=diode voltage, q =electronic charge, ($1.60217662 \times 10^{-19}$ coulombs) K = Boltzmann constant ($1.38064852 \times 10^{-23}$ J/K), T = Temperature in Kelvin.
- ▶ The current is in the range 0.1nA to 0.5nA per device at room temperature.
- ▶ This needs to be compensated by refreshing at regular intervals.
- ▶ There will be some sub-threshold leakage current
- ▶ **Solution** –
- ▶ High V_t transistor
- ▶ Use weak PMOS transistor.

Charge sharing Problem

- In Pre-charging phase load capacitor (C_L) is charge to V_{dd} .
- So the total charge $Q = V_{dd} \cdot C_L$.
- In Evaluation phase if $ABC = 111$ then C_L will discharge to '0'
- If input is $ABC = 110$, then after redistribution of charge the out node will attain the value suppose V_A



Continue.....

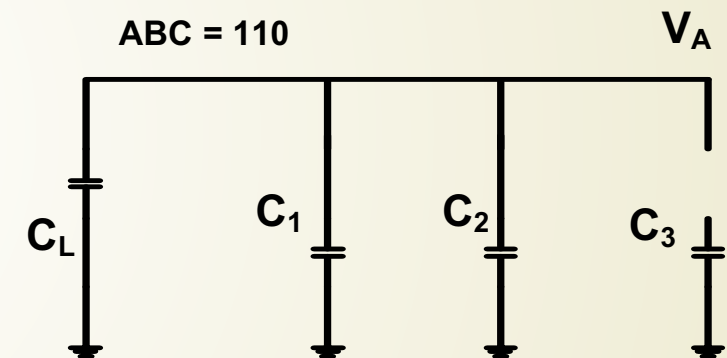
- According to law of conservation , the total charge will remain same , but it get distributed

- $C_L V_{DD} = V_A (C_L + C_1 + C_2)$

- $V_A = \frac{C_L V_{DD}}{C_L + C_1 + C_2}$

- Let $C_1 = C_2 = 0.5 C_L$

- $\frac{C_L V_{DD}}{C_L + 0.5C_L + 0.5C_L} = \frac{C_L V_{DD}}{2C_L} = \frac{V_{DD}}{2}$



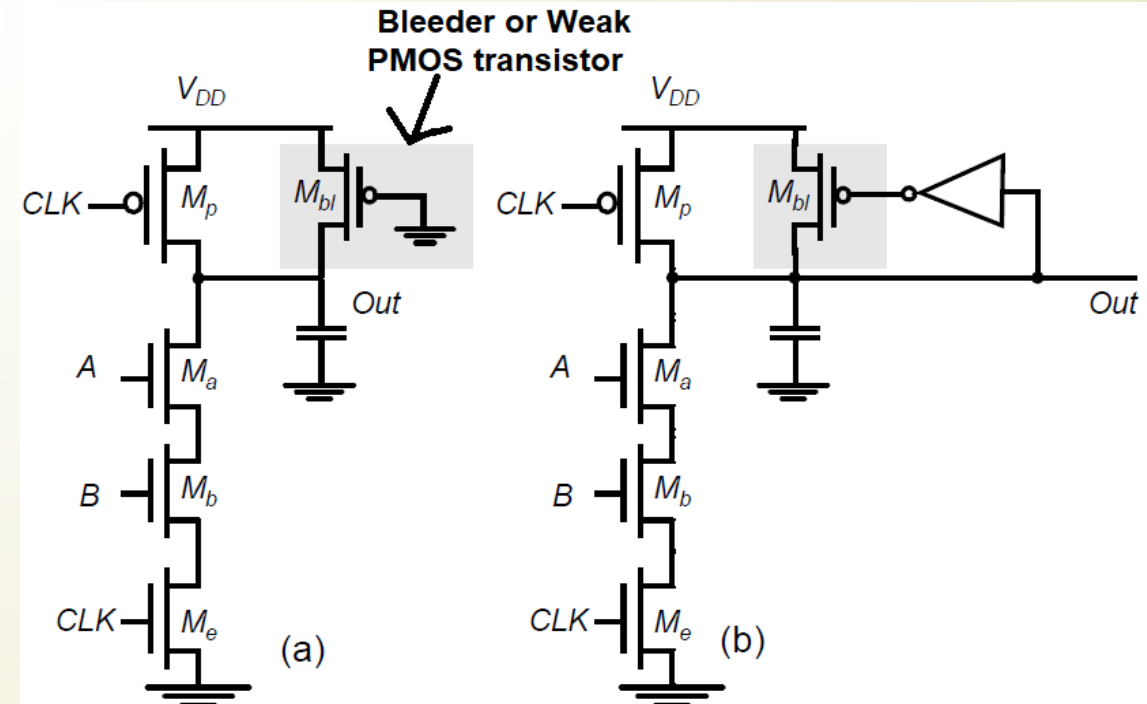
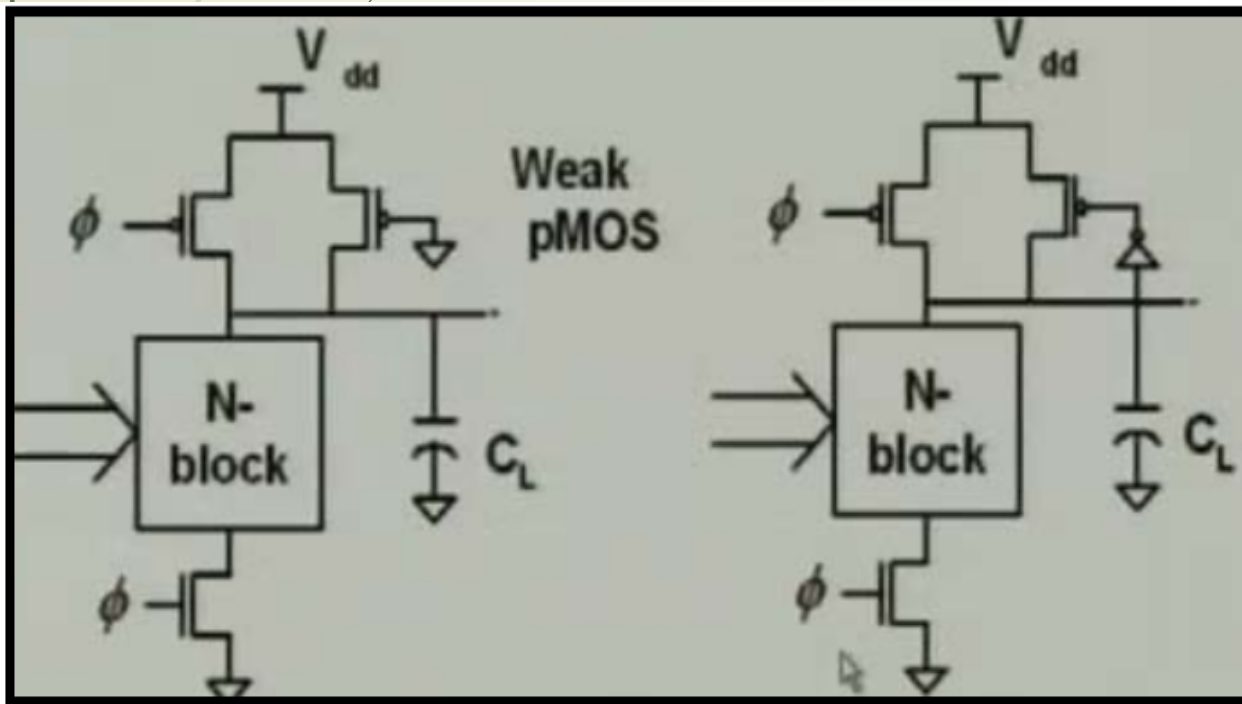
Re-distribution of charges

Continue.....

- ▶ So after distribution the charge remain to $V_{DD}/2$ because of parasitic capacitance present on the internal node.
- ▶ The charge that was present during the pre-charging mode get distributed leading to reduction in output voltage which may be treated as low level. So this problem is called **charge sharing problem**.

How to Overcome charge leakage and sharing problem

- ▶ This problem can be overcome by using a weak pMOS transistor (bleeder transistor). This is also used to overcome the problem of charge leakage problem. So, a weak pMOS is added as pull-up transistor, this transistor always remains on and behaves like a pseudo n-MOS circuit during evaluation period.
- ▶ by weak I mean that resistance is very high. In other words the current that it will supply is very small, when the resistance is high current that will be, it will supply to the load capacitance will be small, and as a consequence it will replenish the charge that is lost; either due to leakage problem or due to charge sharing problem.



Capacitive Coupling

- ▶ The high impedance of the output node makes the circuit very sensitive to crosstalk effects. A wire routed over a dynamic node may couple capacitively and destroy the state of the floating node.
- ▶ A transition in the input In of the static gate may cause the output of the gate (Out_2) to go low.

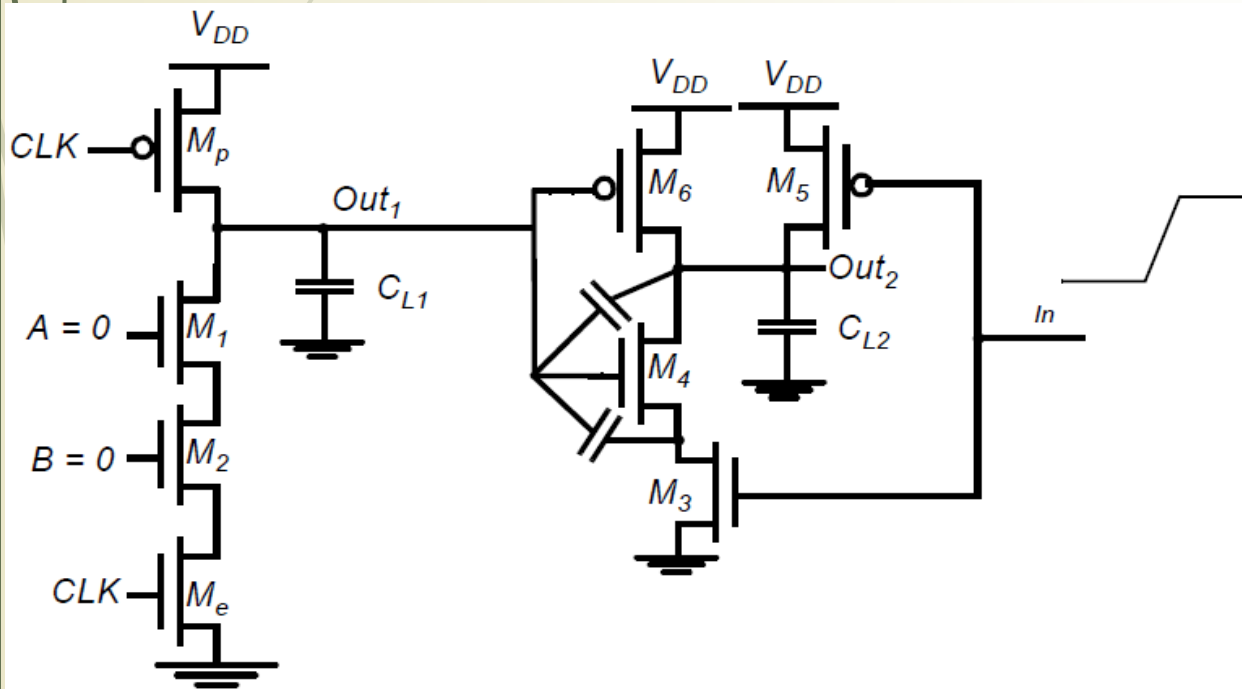


Figure Example demonstrating the effect of backgate coupling

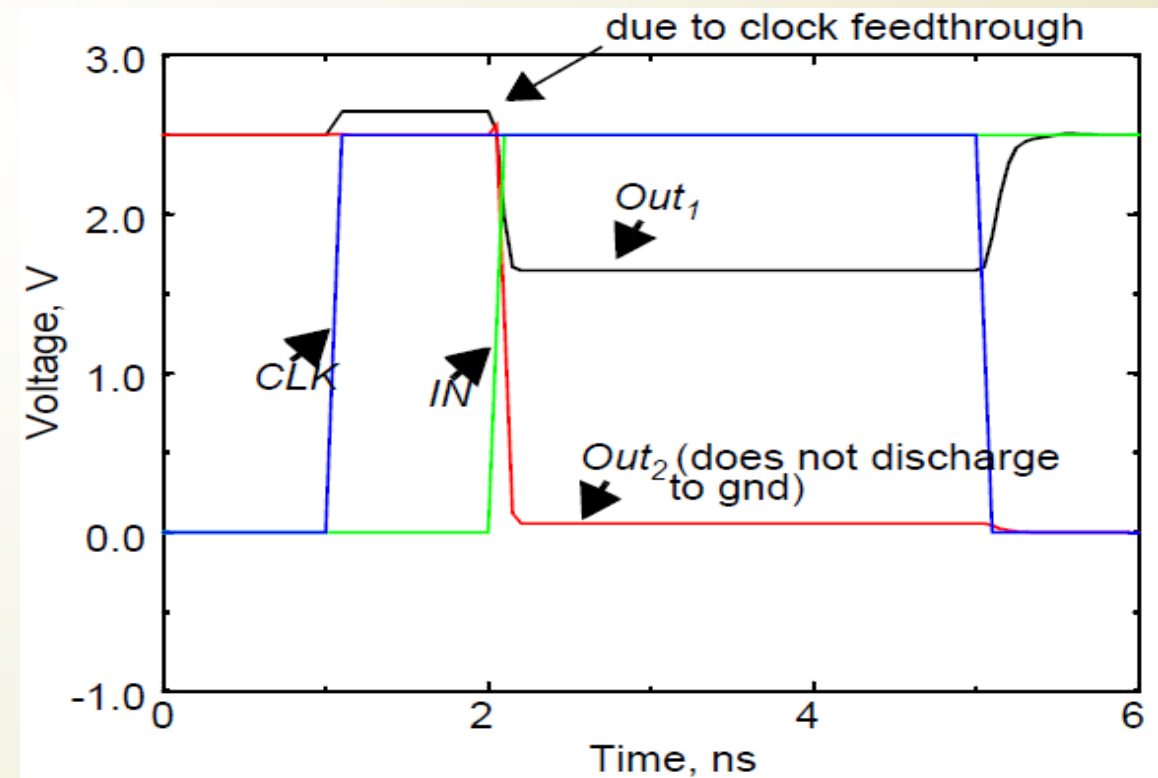


Figure Clock feedthrough effect.

Continue....

- This output transition couples capacitively to the other input of the gate, the dynamic node Out1, through the gate-source and gate-drain capacitances of transistor M4.
- This further causes the output of the static NAND gate not to drop all the way down to 0V. Consequently, the circuit can evaluate incorrectly.

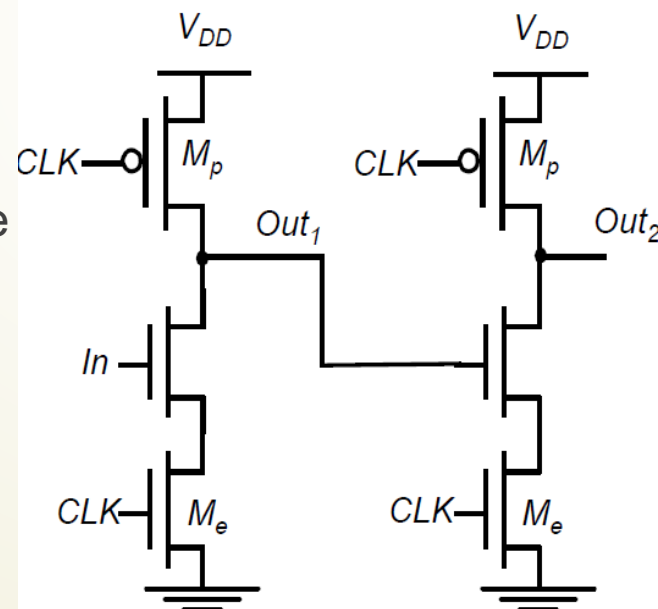
➤ Clock-Feedthrough

- A special case of capacitive coupling is clock-feedthrough, an effect caused by the capacitive coupling between the clock input of the precharge device and the dynamic output node.
- This capacitive coupling causes the output of the dynamic node to rise above V_{DD} on the low-to-high transition of the clock, assuming that the pull-down network is turned off. Sometime they may lead to faulty operation.
- All the above considerations demonstrate that the design of dynamic circuits is rather tricky and requires extreme care.

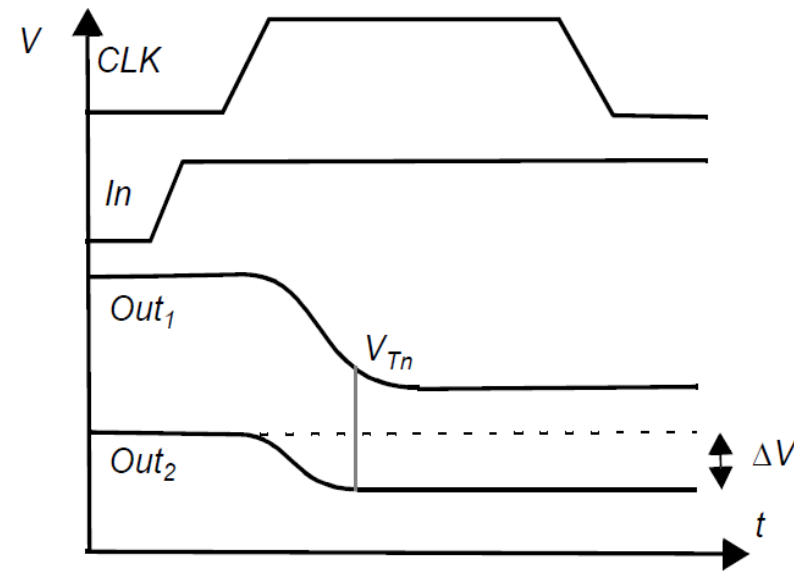
Cascading Dynamic Gates

- Besides the signal integrity issues, there is one major catch that complicates the design of cascaded dynamic circuits.
- During the precharge phase (i.e., $CLK = 0$), the outputs of both inverters are precharged to V_{DD} . Assume that the primary input In makes a $0 \rightarrow 1$ transition (Figure 6.64b). On the rising edge of the clock output Out_1 starts to discharge.
- The second output should remain in the precharged state of V_{DD} as its expected value is 1.
- There is a finite propagation delay for the input to discharge Out_1 to GND.
- Therefore, the second output also starts to discharge. Because of this precious charge is lost.

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(a)



(b)

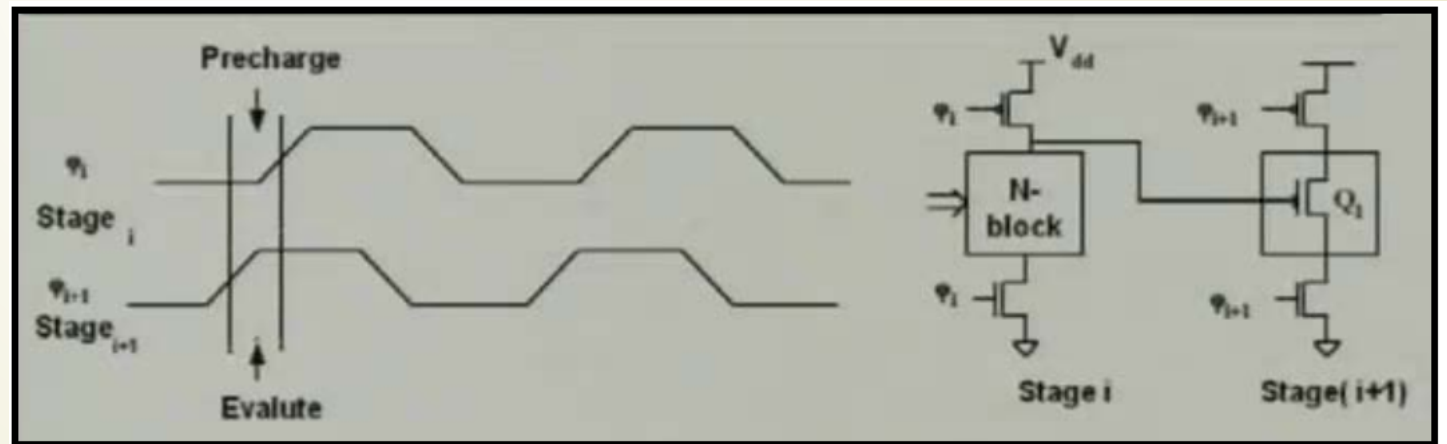
Figure Cascade of dynamic n -type blocks.

Continue....

- This leaves Out2 at an intermediate voltage level. The correct level will not be recovered. The charge loss leads to reduced noise margins and potential malfunctioning.

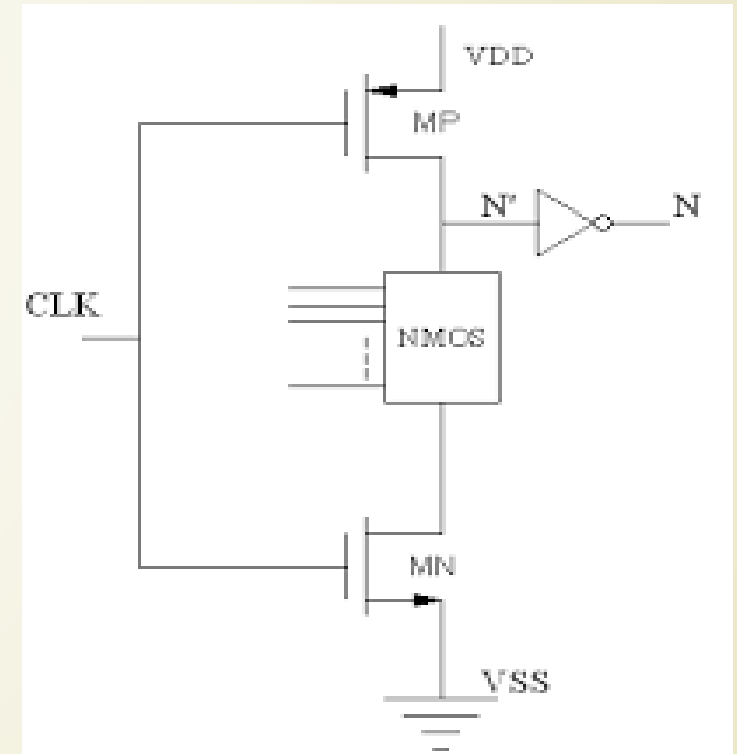
Clock Skew Problem

- ▶ Clock skew problem arises in dynamic CMOS circuits, particularly whenever you are realizing multilevel circuits.
- ▶ Clock skew problem arises, because you have to distribute the clock throughout the entire circuit, but the depending on the clock tri-network, the clock signal can reach different points of the circuit at different points of the time, because of different amounts of delay of the clock tree network.
- ▶ Furthermore, this delay due to resistance and parasitic capacitance associated with the wire. It result in hazard and race condition (or incorrect output).
- ▶ This problem can be overcome by setting the output low during pr-charge.



Domino CMOS Circuits

- ▶ The problem with faulty discharge of Pre-charged nodes in CMOS dynamic logic circuits can be solved by placing an inverter in series with the output of each gate.
- ▶ Domino gate consist of two distinct components:
 - The first component is a conventional nMOS gate.
 - The second component is a static inverting CMOS buffer.
- For example, during pre-charge you are applying a inverter here. So, when it is high, then this is low.



continue.....

So, this cannot really discharge and then you can realize the circuit, it will always get correct output irrespective of this delay in irrespective of this clock skew problem. so; that means, even if the clock arises late to the first stage, it will not lead to premature discharge of this capacitor during evaluation phase of the stage and pre-charge phase of the previous stage.

The introduction of the static inverter has the additional advantage that the fan-out of the gate is driven by a static inverter with a low impedance output, which increases noise immunity. The buffer furthermore reduces the capacitance of the dynamic output node by separating internal and load capacitances

Advantages

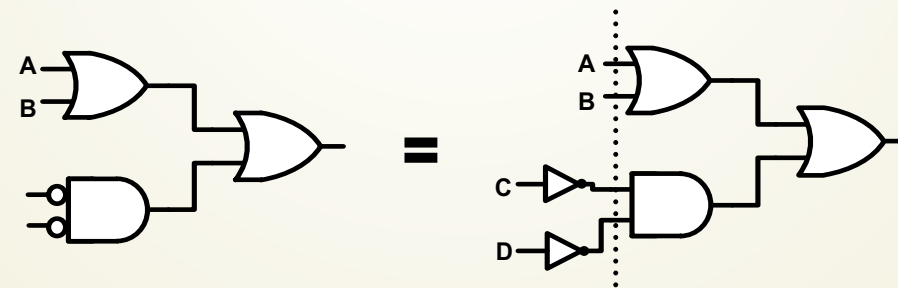
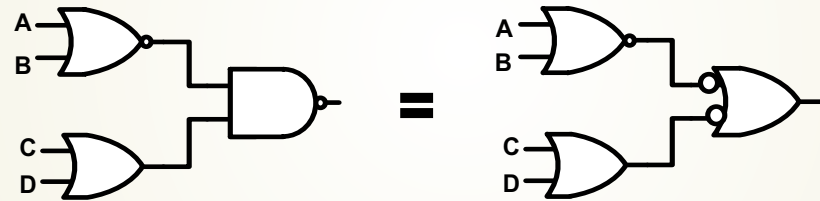
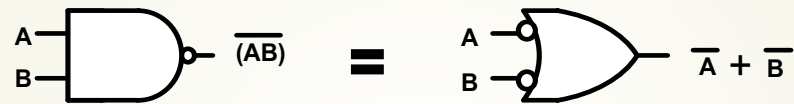
- Low power consumption
- Reduced chip area
- Higher speed of operation
- No short circuit power dissipation
- No glitching power dissipation

Limitations:

- Each gate requires an inverter buffer
- All gates are non-inverting in nature
- Higher switching activity
- How to deal with non-inverting property of domino logic ?
- Re-organize the logic using bubble shifting to transform the circuit.
- Dual rail domino logic.

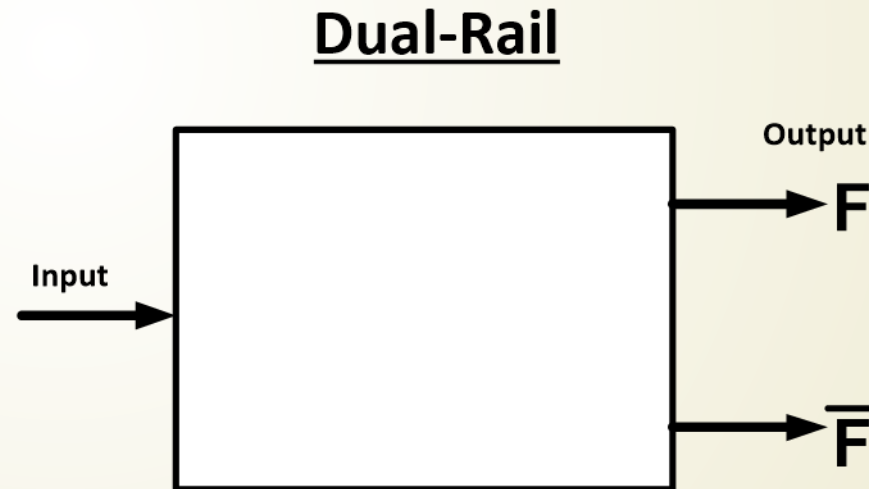
Bubble Shifting

Bubble Shifting

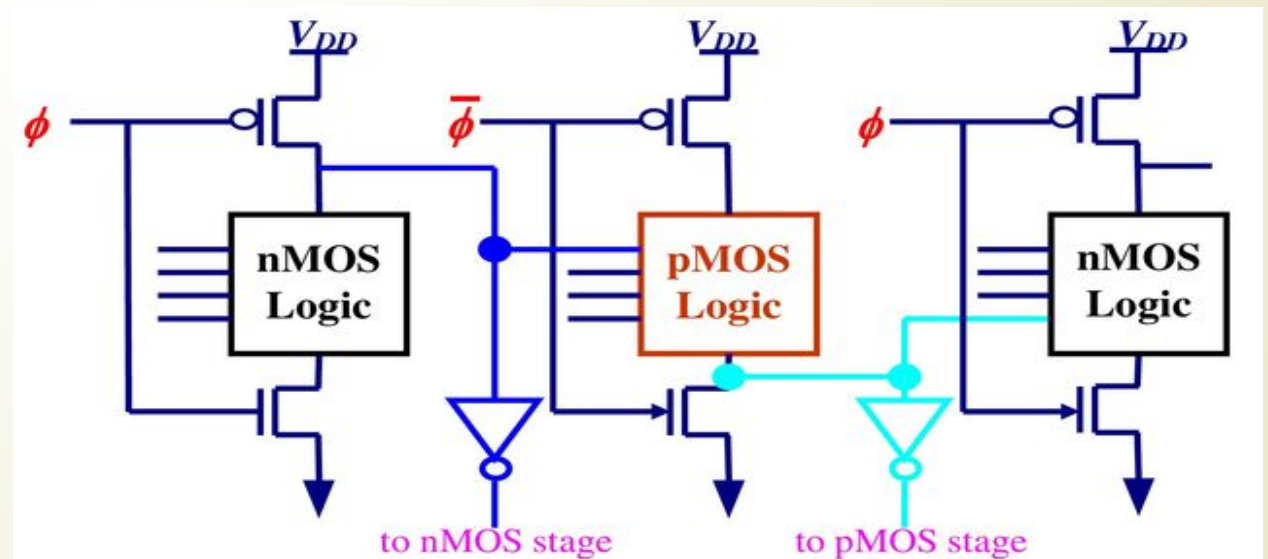


Dual Rail

- we are applying input and producing F , in addition to F you will be also realizing $\bar{F}$; that means, you are converting a dual rail circuit; that means, basic domino circuit can be modified to realize F as well as $\bar{F}$ simultaneously. Then it will be you know wherever inverter is required will be applying inverter. So, it is no longer realizing only positive unit function, because it is realizing positive unit and its compliment



- NORA stands for No Race
- By alternatively using p-blocks and n-blocks, the clock skew problem is overcome in NORA logic circuits.
- we have seen that the, when during pre-charge the previous stage is disturbing the next stage if it is in the evaluation phase. Now, if it is a pMOS transistor, then a high level output will not disturb it. So, what we are doing, we are using alternatively a n-MOS block followed by pMOS block .

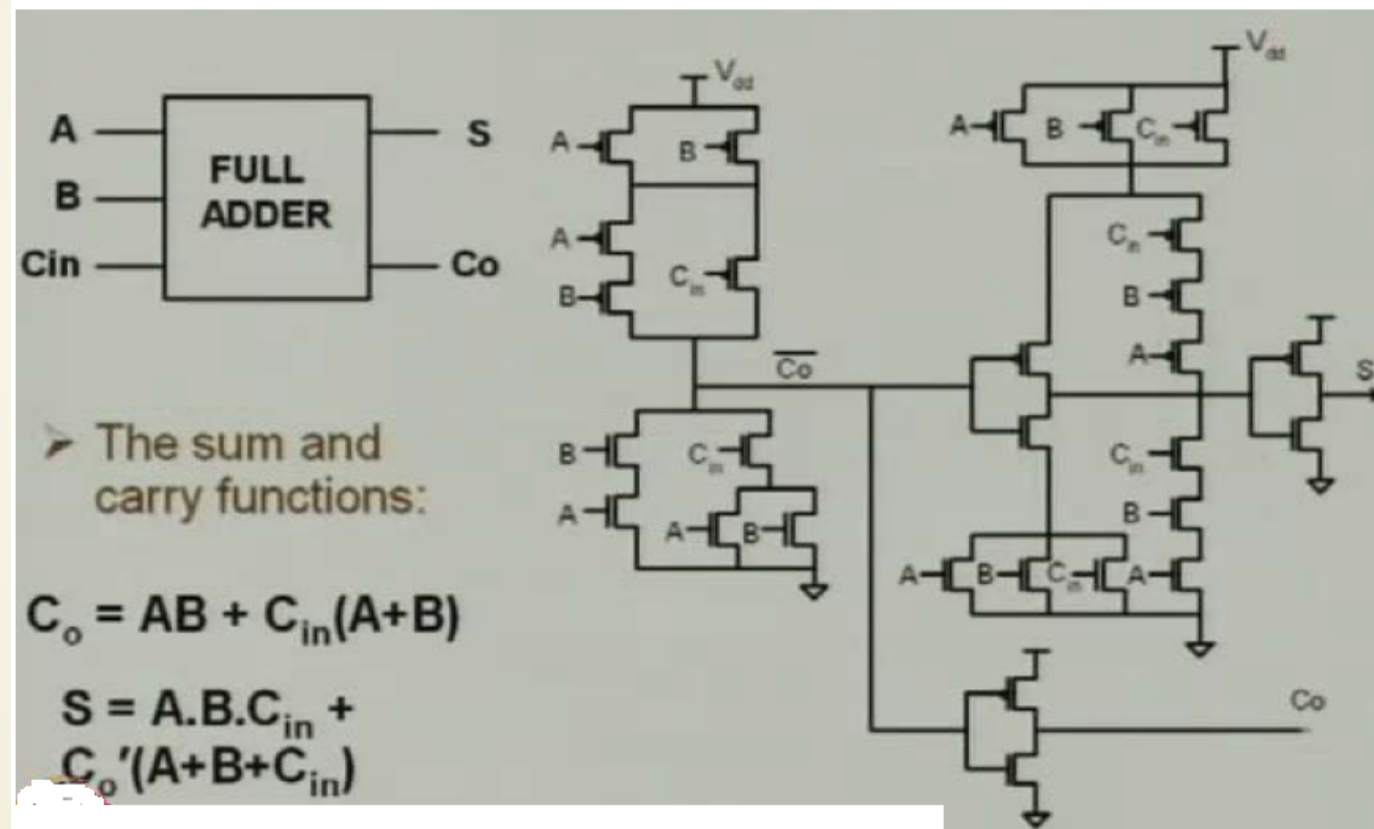




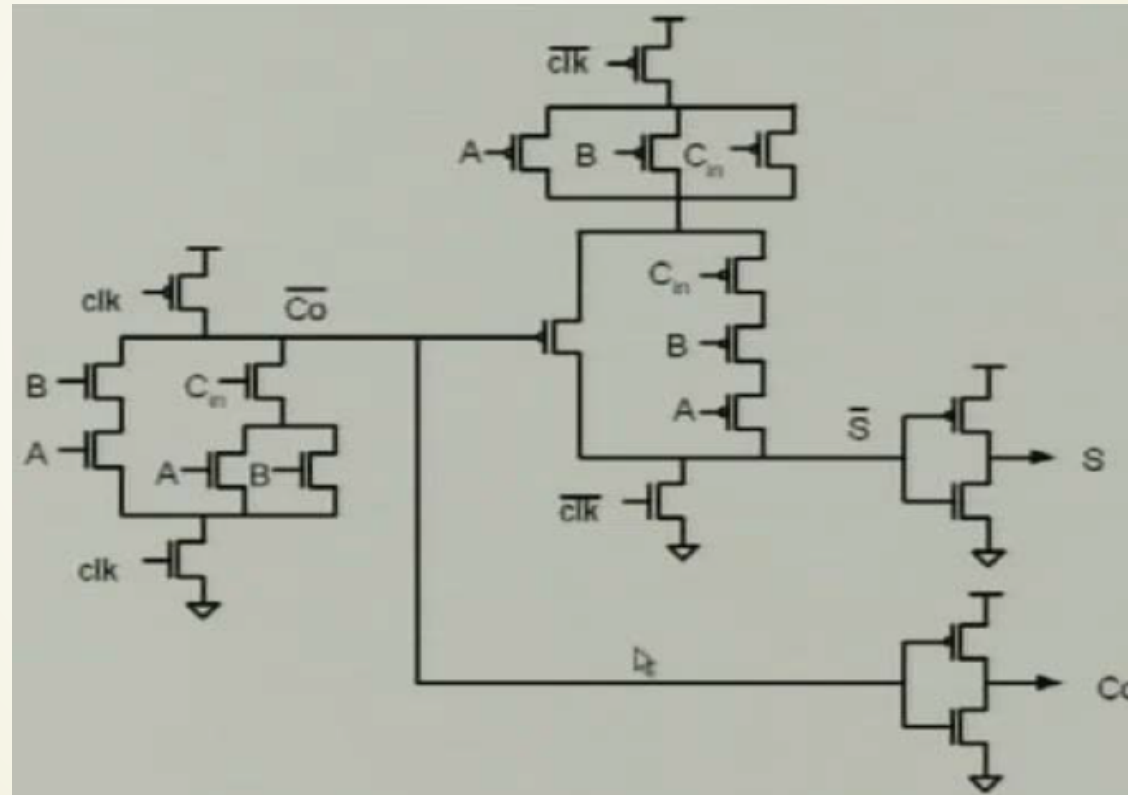
Continue....

- ▶ Compared to domino technique, this approach gives higher flexibility.
- ▶ Both inverted and non-inverted signals are available from NORA- technique.
- ▶ The DEC-alpha processor, the first 250 MHz CMOS microprocessor, made extensive use of NORA logic Circuits.

Static CMOS full - Adder



NORA CMOS realization of full adder circuit





Thank You

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